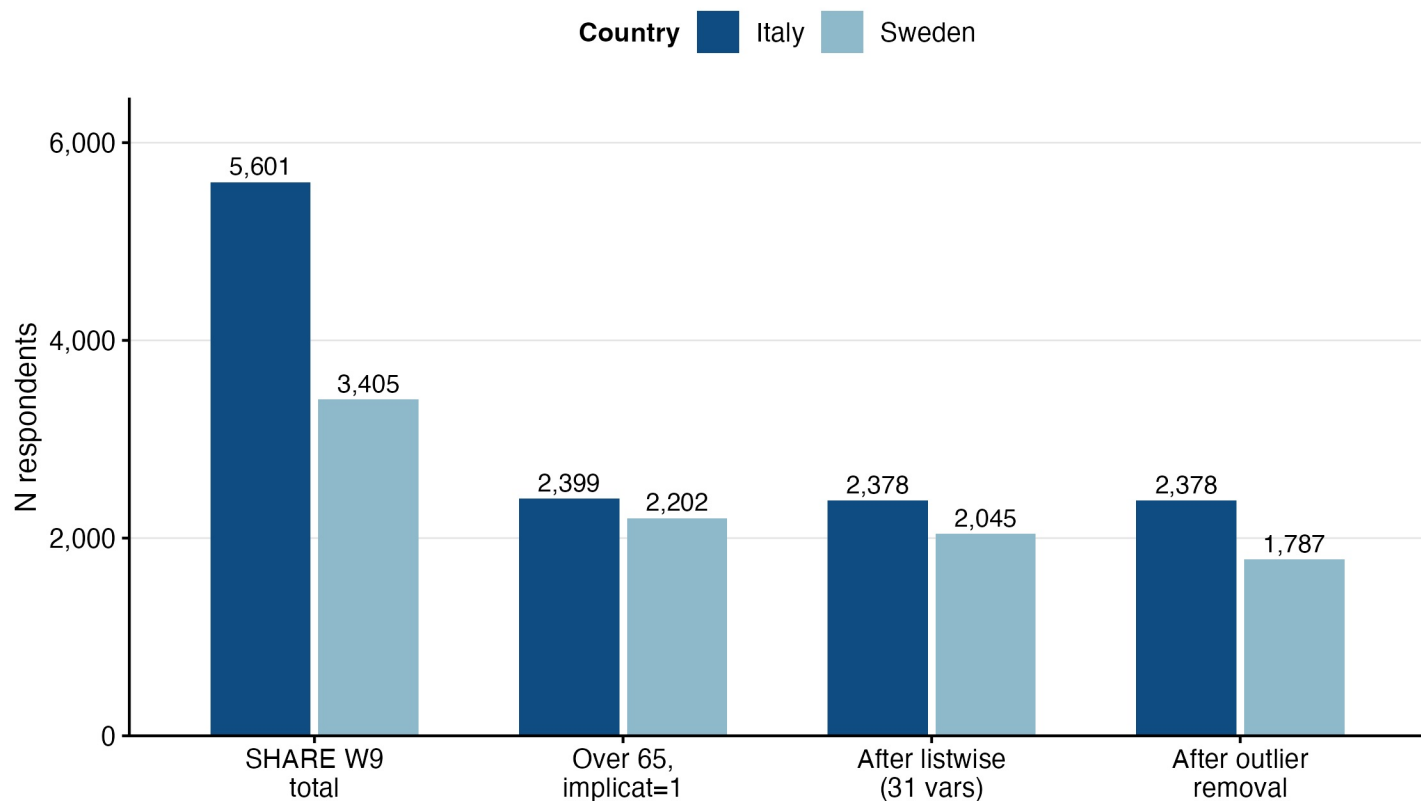


Figure 1. Analytical sample derivation, SHARE Wave 9

Italy and Sweden cohorts, from survey total to final pre-clustering sample

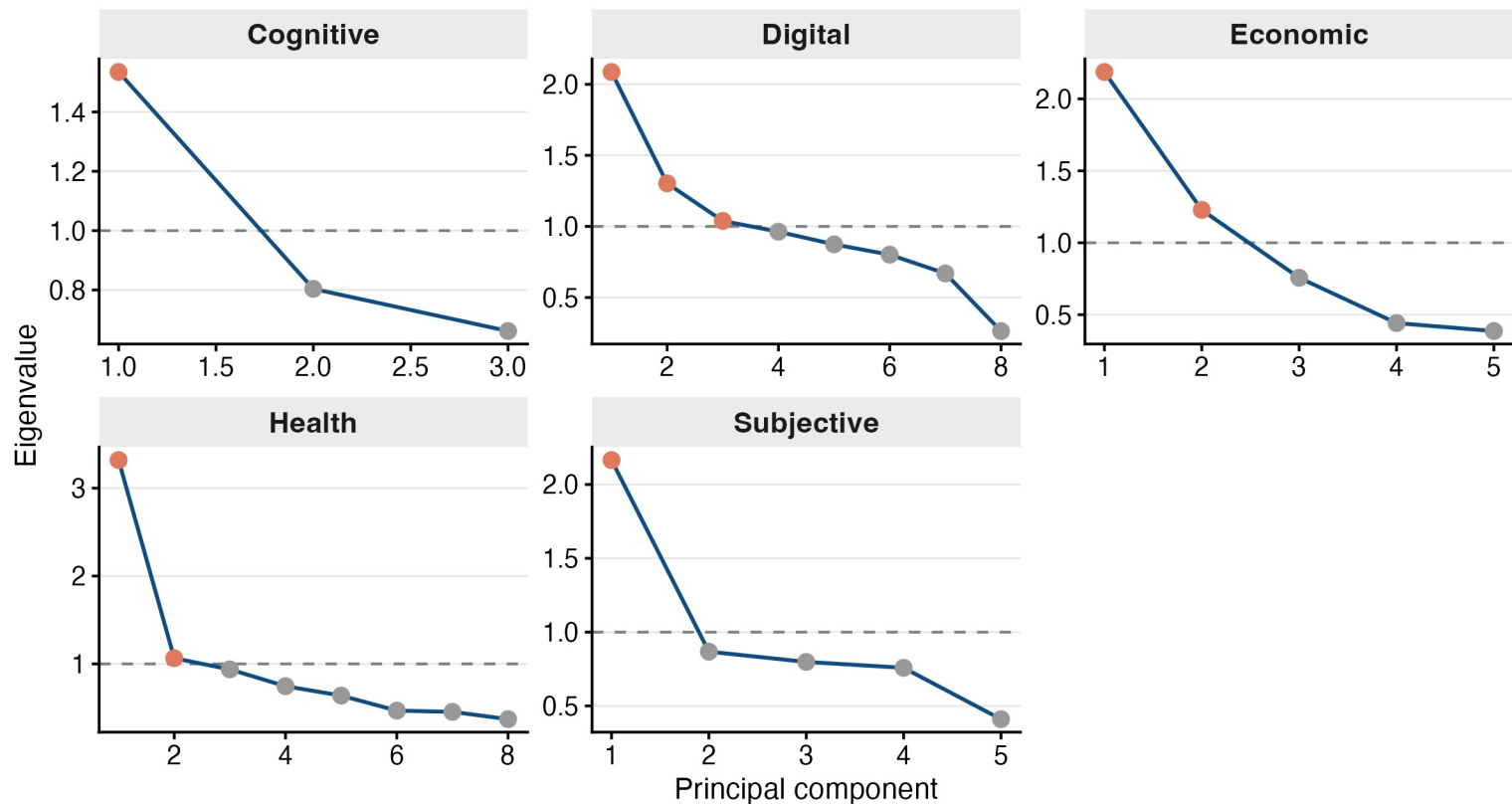


Source: SHARE W9, own calculations.

Figure 2. PCA scree plots by dimension, Italy

Eigenvalues per principal component; dashed line = Kaiser criterion (eigenvalue = 1)

● Below threshold ● Kaiser (eigenvalue > 1)

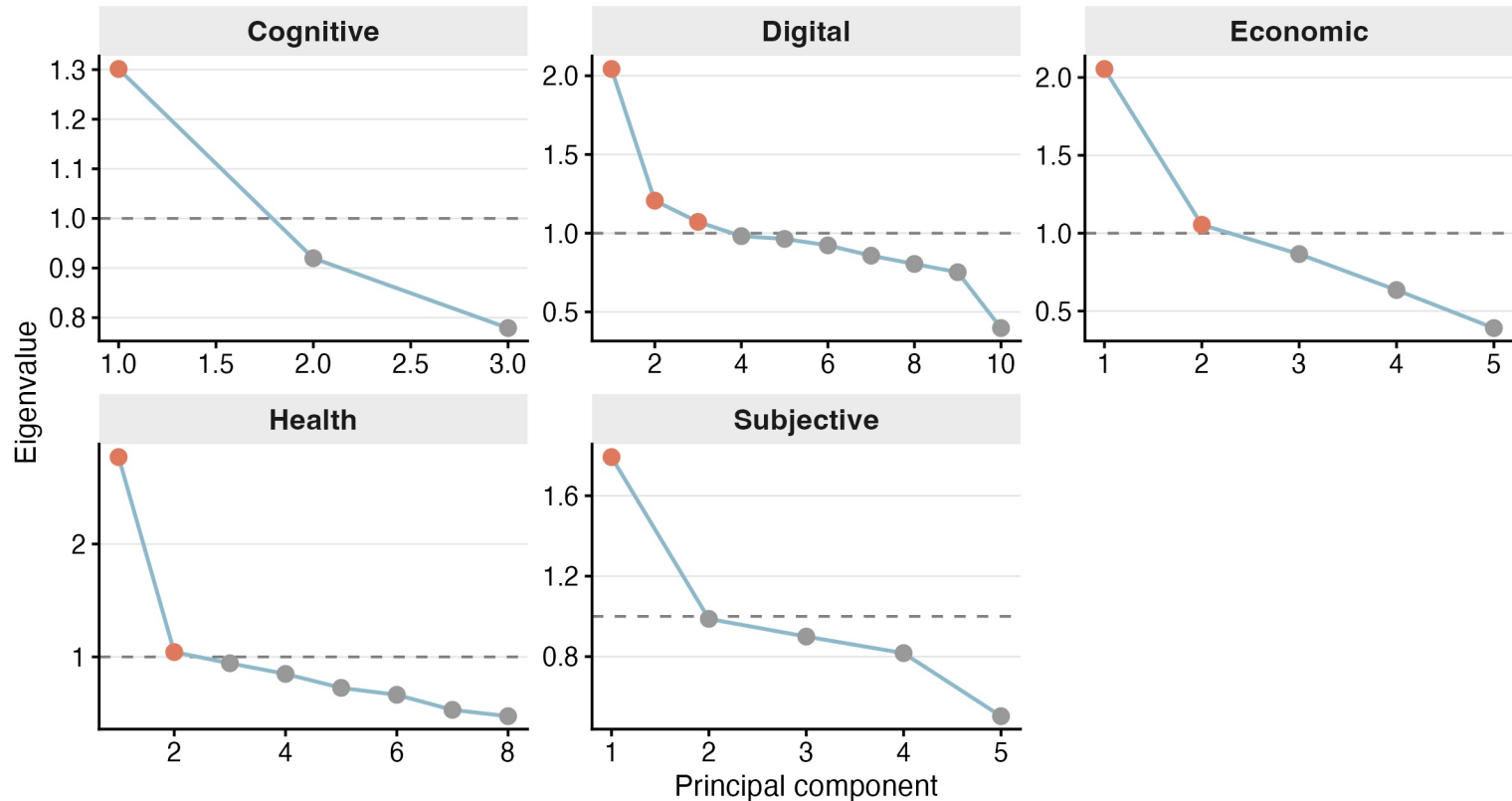


Italy n=2,378. Digital/Social dimension excludes ac035d4 and ac035d7 (zero variance).

Figure 3. PCA scree plots by dimension, Sweden

Eigenvalues per principal component; dashed line = Kaiser criterion (eigenvalue = 1)

● Below threshold ● Kaiser (eigenvalue > 1)



Sweden n=1,787. All 31 variables active.

Figure 4. Factor loadings matrix, Italy (PA extraction, varimax rotation)

Six factors over 29 active variables; values shown if |loading| >= 0.30



Italy n=2,378. KMO=0.832, Bartlett p<0.001. 6 factors explain 38.4% variance.

Figure 5. Factor loadings matrix, Sweden (PA extraction, varimax rotation)

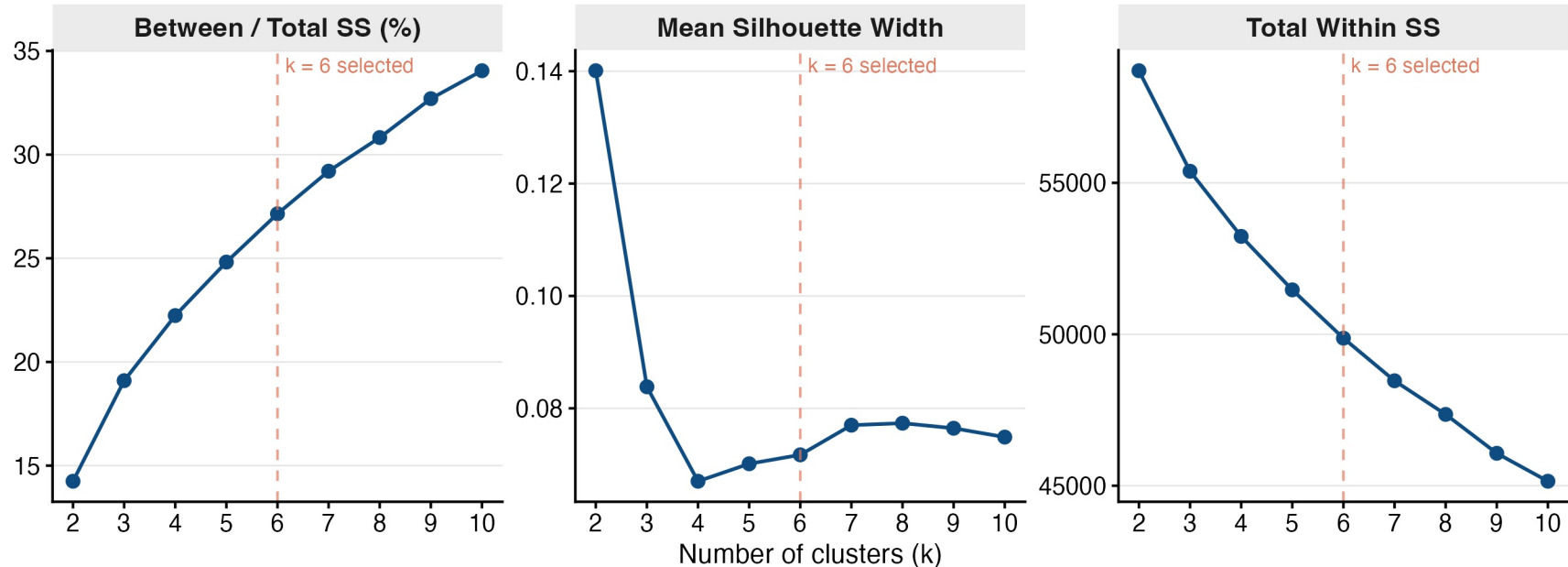
Six factors over 31 active variables; Heywood case on social_integration (loading > 1)



Sweden n=1,787. KMO=0.808. social_integration h²=1.38 (Heywood).

Figure 6. K-means diagnostics across k, Italy

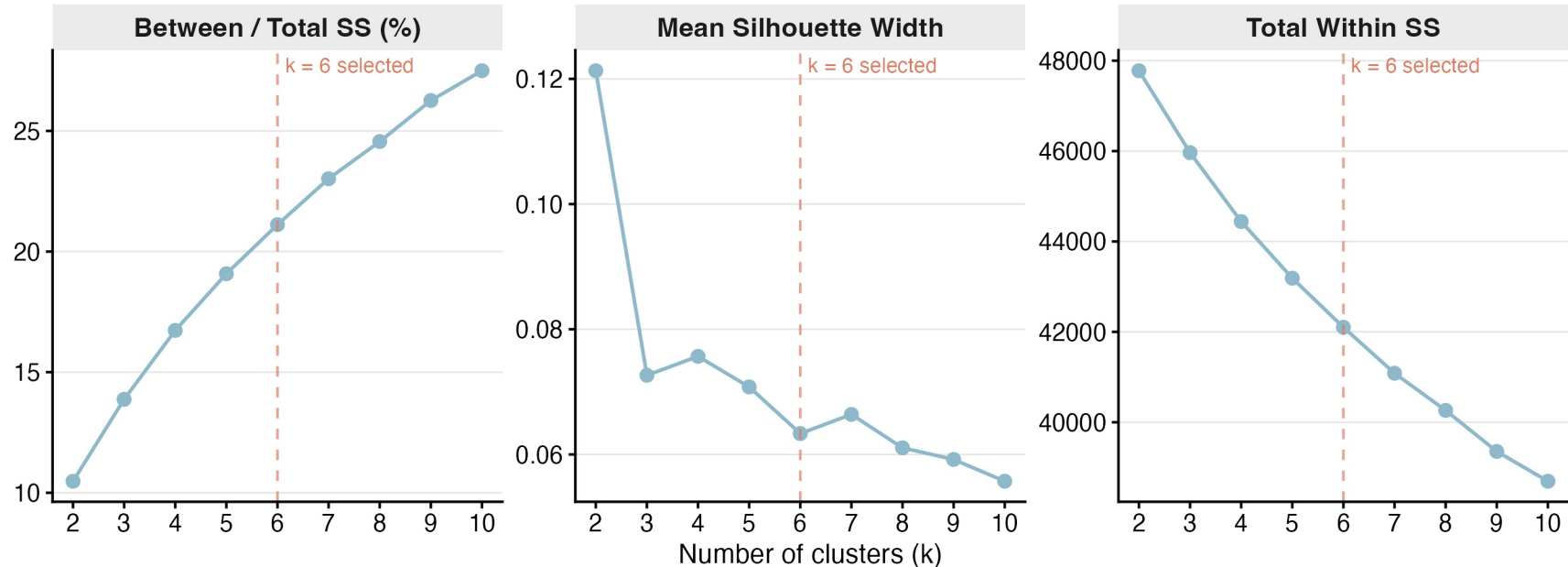
Elbow (Total Within SS), separation ratio, and silhouette width



Italy n=2,378; k=6 chosen for cross-country comparability (no clear elbow at k<6).

Figure 7. K-means diagnostics across k, Sweden

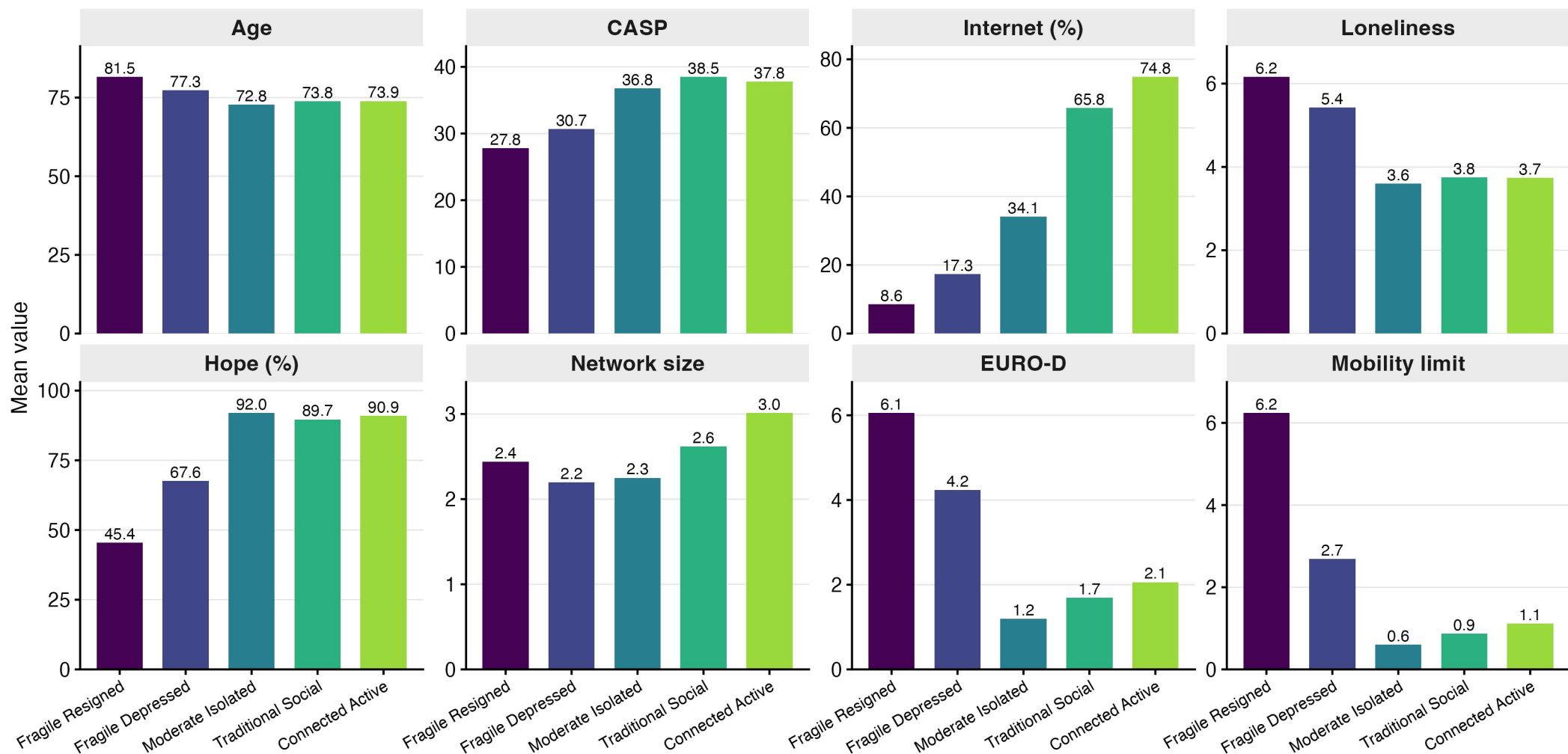
Elbow (Total Within SS), separation ratio, and silhouette width



Sweden n=1,787; k=6 chosen for cross-country comparability.

Figure 8. Profile characteristics, Italy (five k-means profiles)

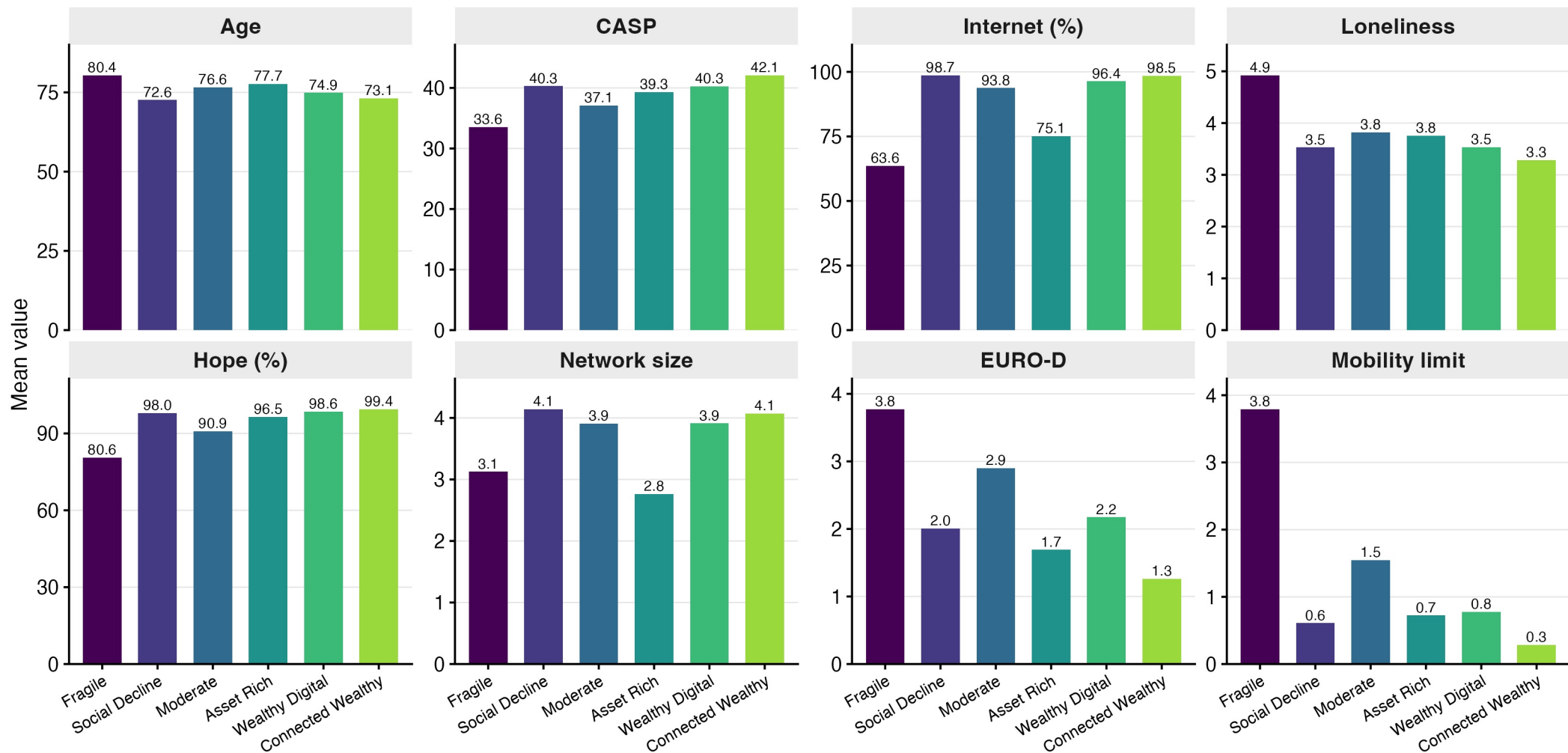
Mean values for each profile on key dimensions of wellbeing and connectedness



Italy n=2,378. Ordered from most fragile (left) to most connected (right).

Figure 9. Profile characteristics, Sweden (six k-means profiles)

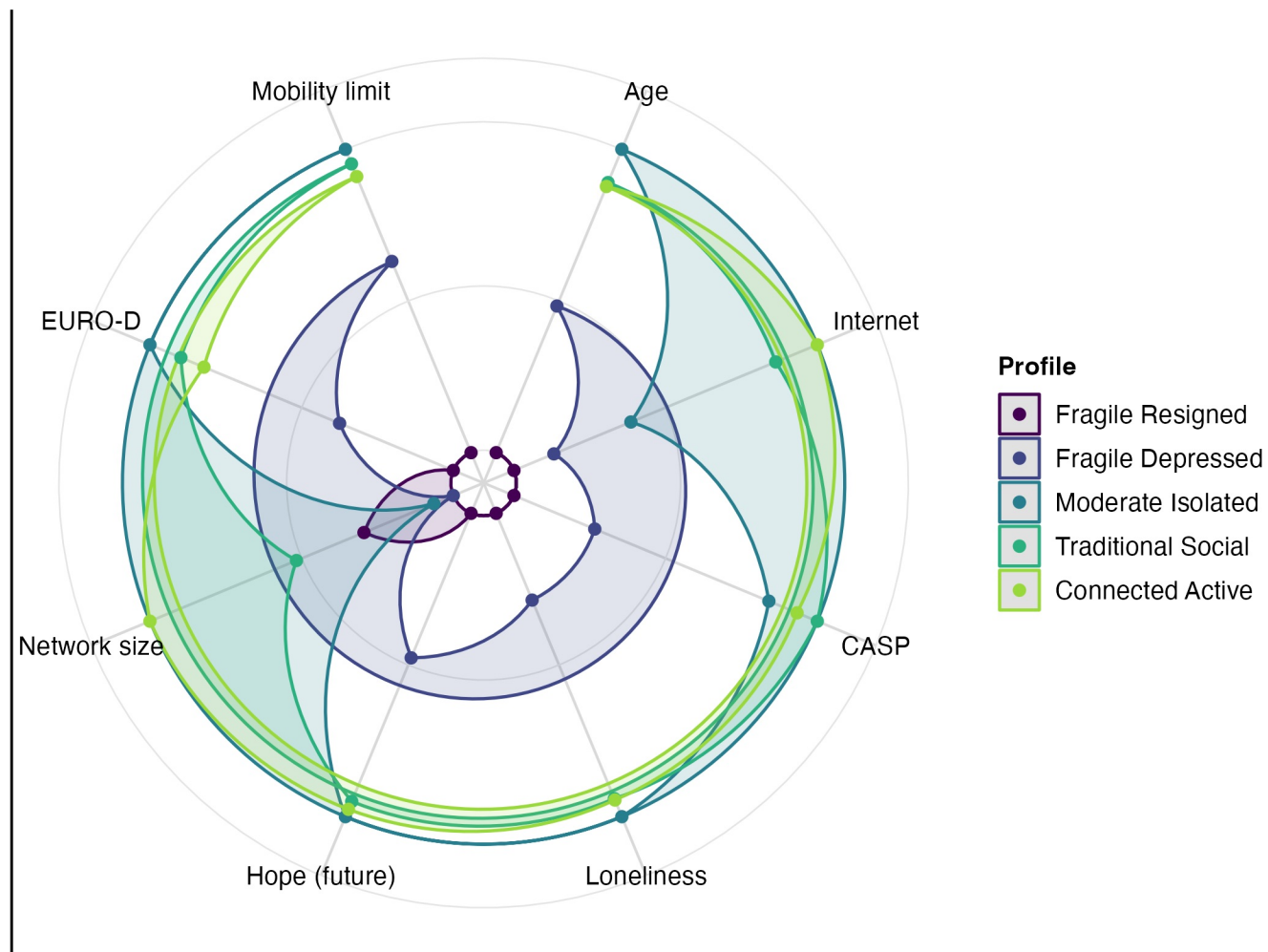
Mean values for each profile on key dimensions of wellbeing and connectedness



Sweden n=1,787. Ordered from most fragile (left) to most connected (right).

Figure 10. Profile radar, Italy

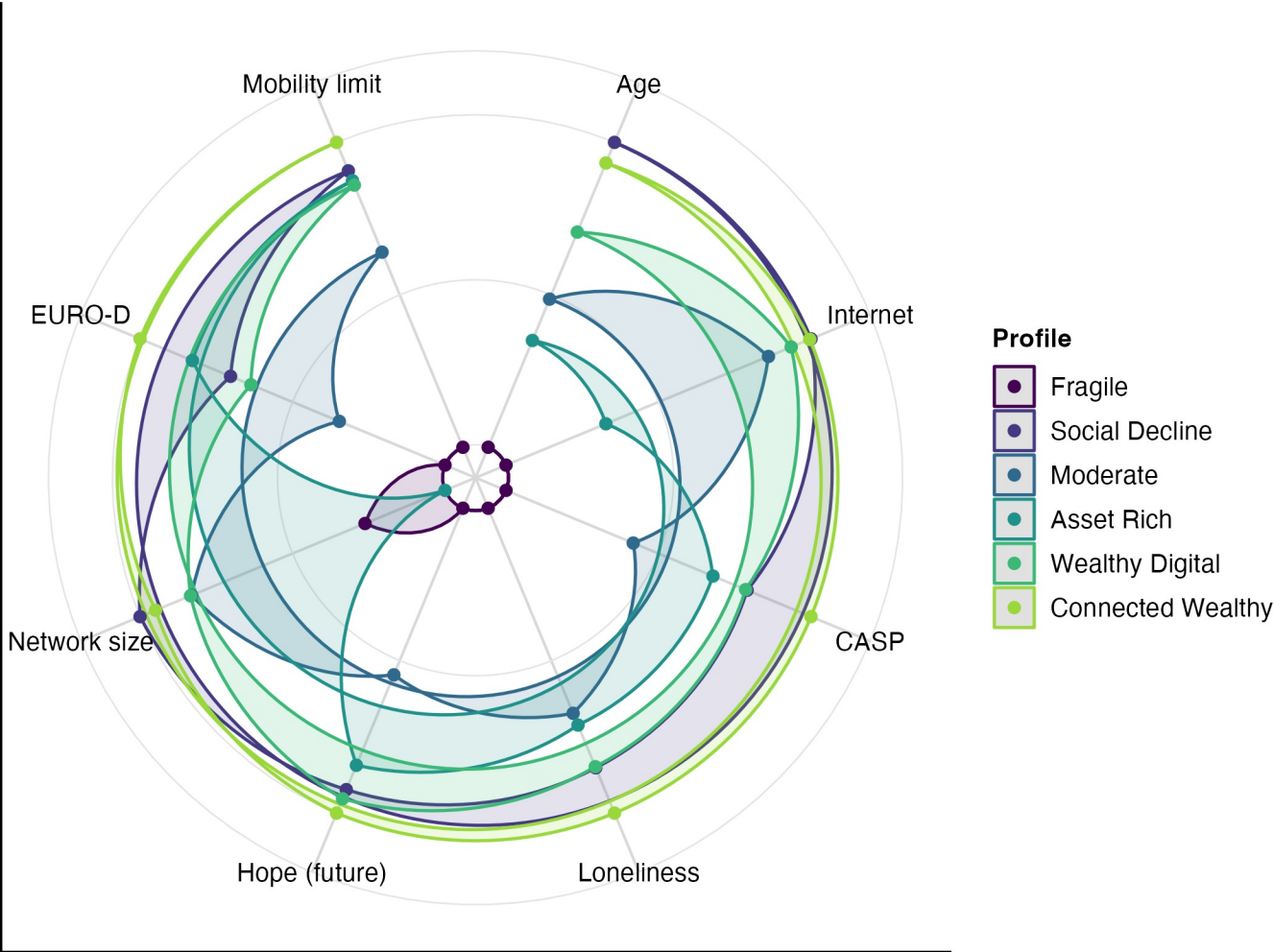
Eight dimensions, min-max normalized (higher = better outcome)



Loneliness, EURO-D, Mobility limit, and Age inverted so higher values denote better outcomes.

Figure 11. Profile radar, Sweden

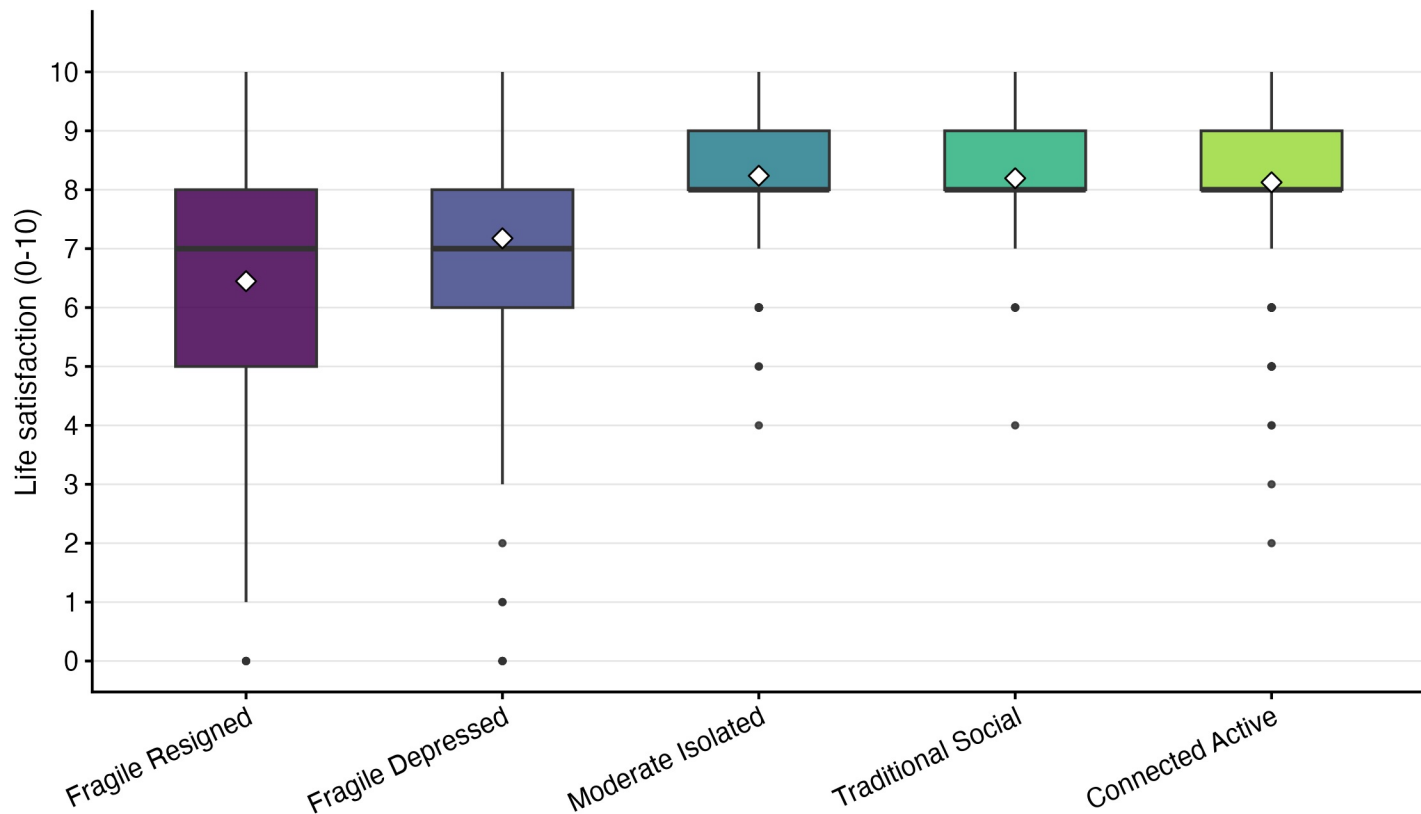
Eight dimensions, min-max normalized (higher = better outcome)



Loneliness, EURO-D, Mobility limit, and Age inverted so higher values denote better outcomes.

Figure 12. Life satisfaction by profile, Italy

One-way ANOVA: $F(4, 2183) = 124.6, p < .001$



Italy n=2,188 (with lifesat non-NA). Diamond = group mean.

Figure 13. Life satisfaction by profile, Sweden

One-way ANOVA: $F(5, 1779) = 44.3, p < .001$

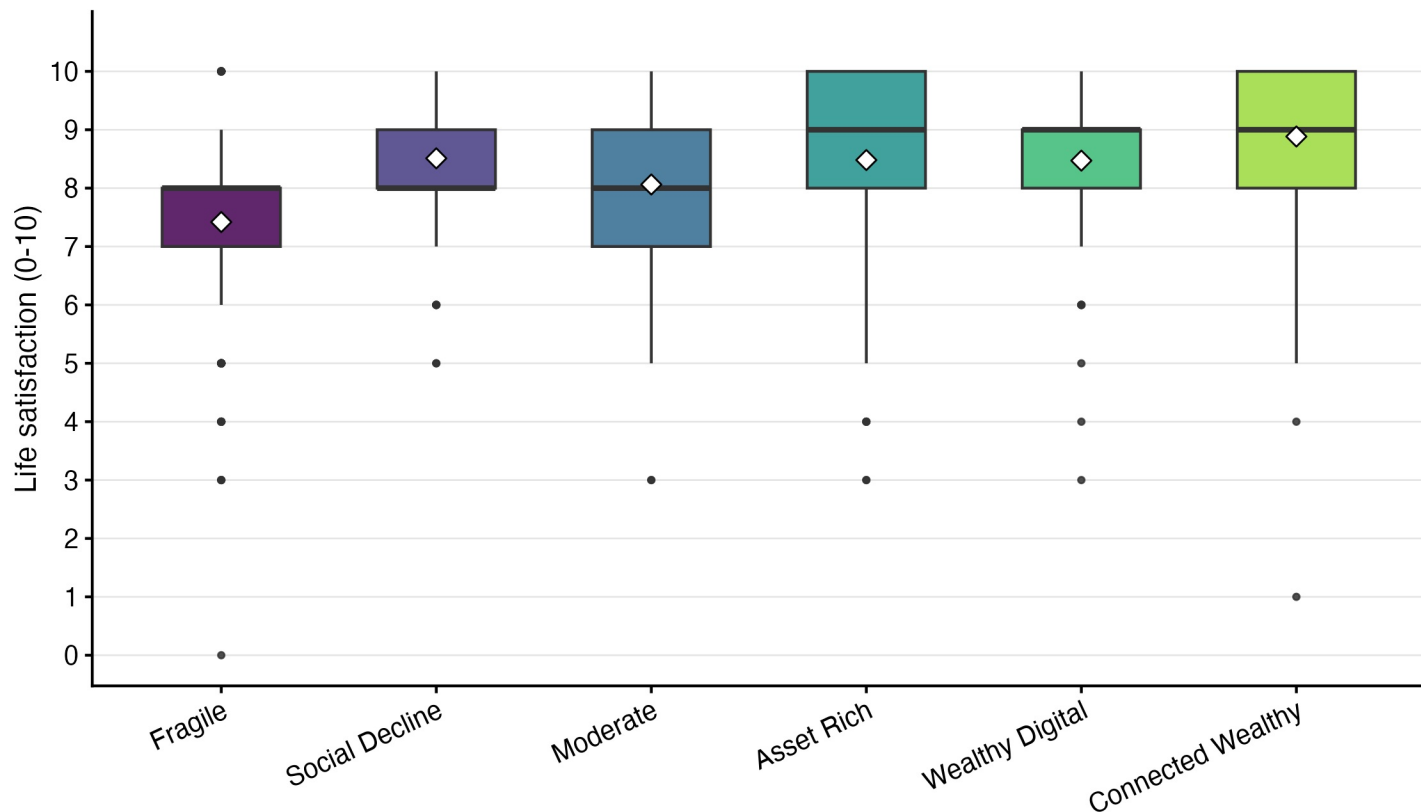
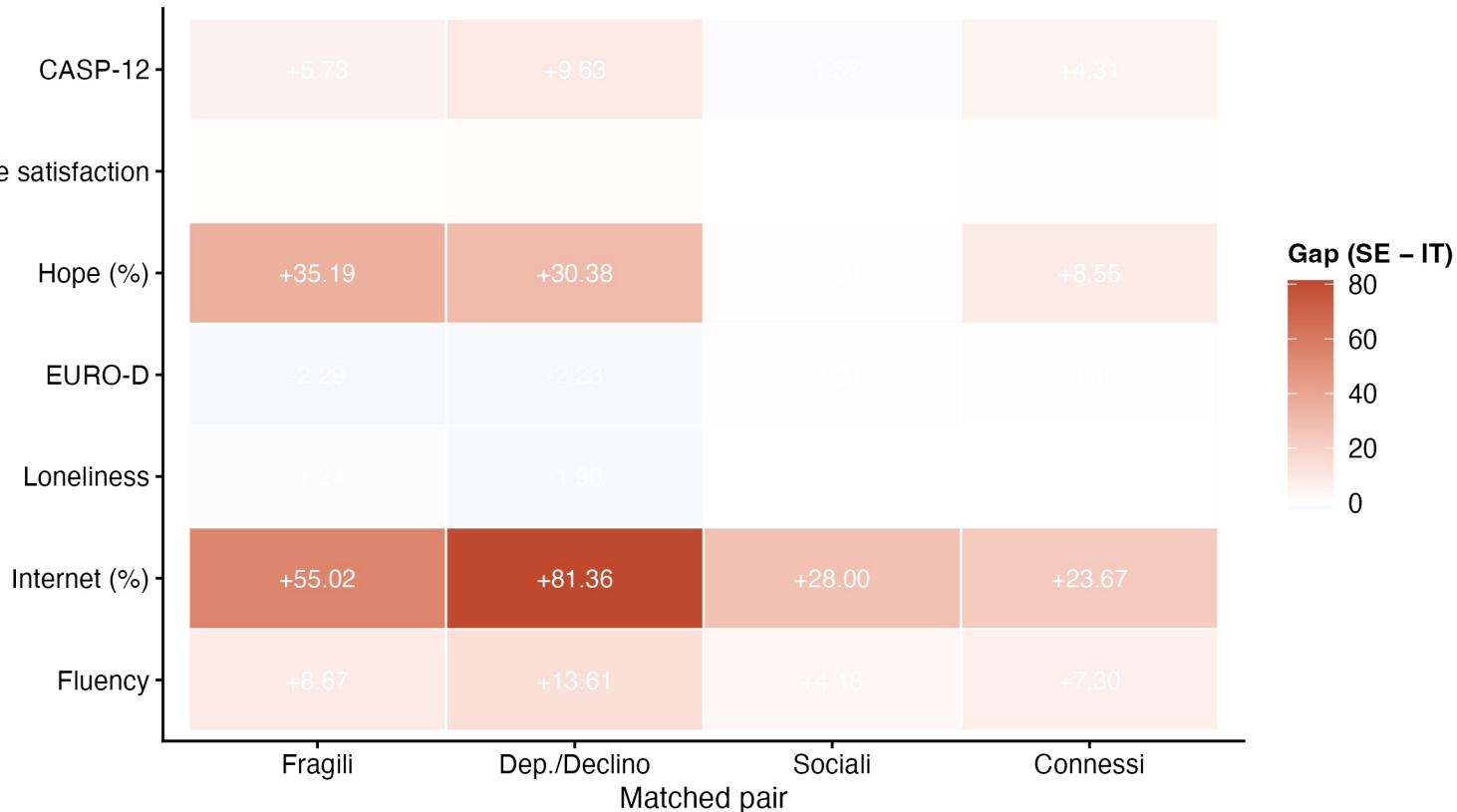


Figure 14. Cross-country gaps across matched profile pairs

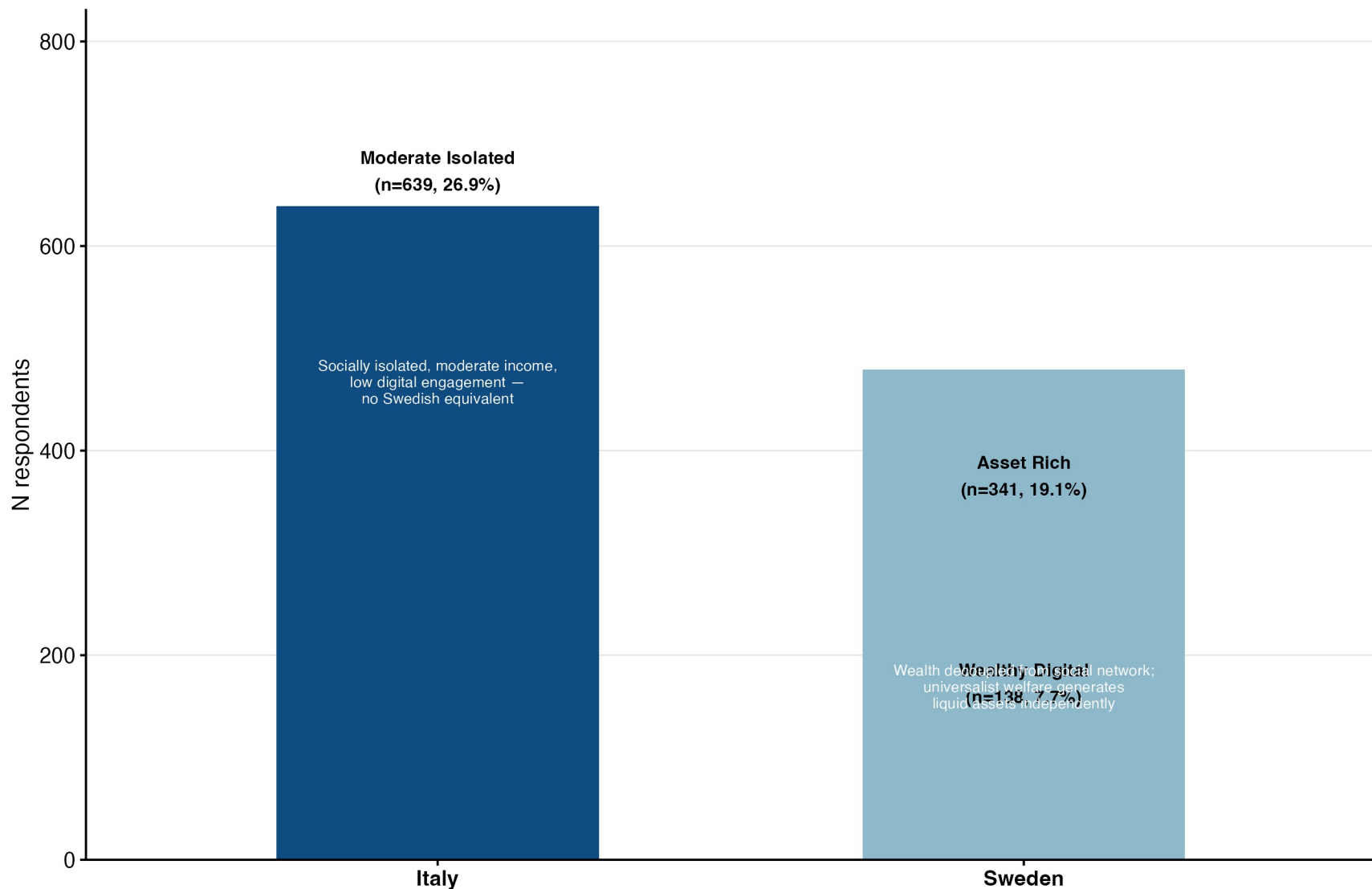
Difference Sweden minus Italy for each wellbeing and connectedness variable



Positive values: higher in Sweden. Gap computed on profile means; Welch t-tests in Chapter 5.

Figure 15. Country-unique profiles and welfare regime specificity

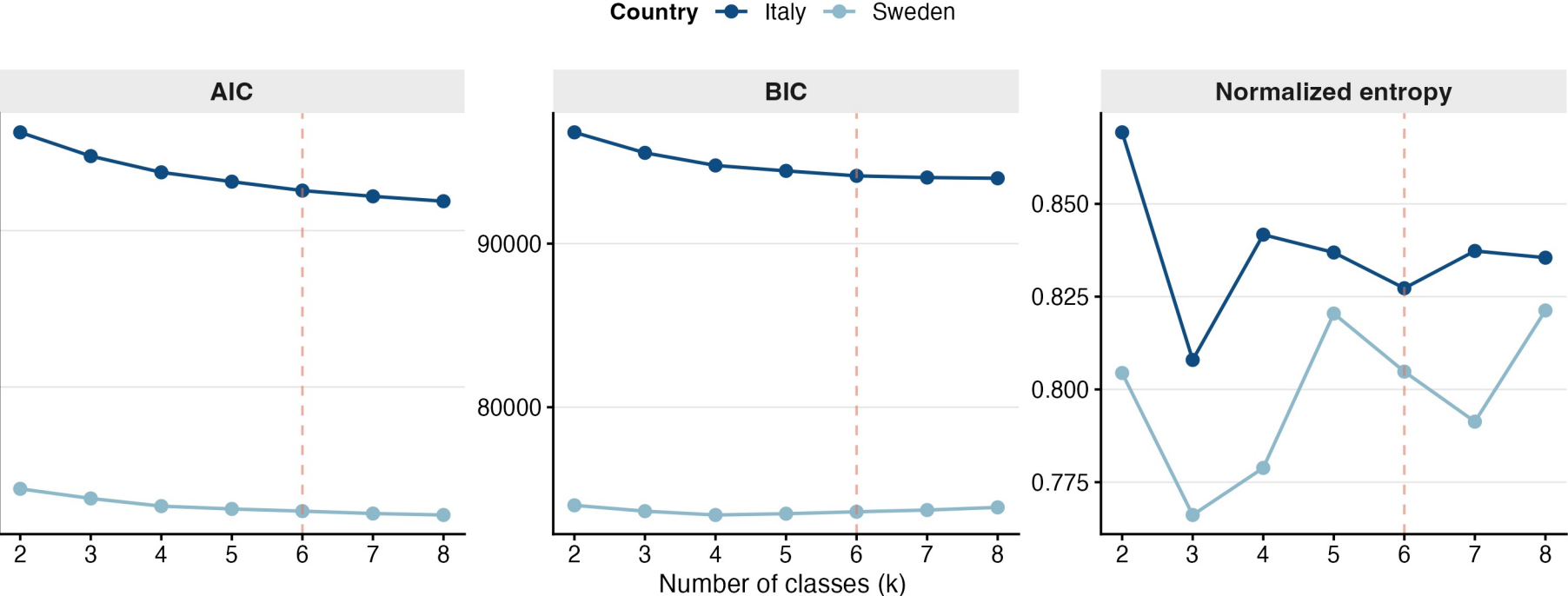
Profiles that emerge in one country but have no analogue in the other



Moderate Isolated (IT) reflects familistic isolation; Asset Rich and Wealthy Digital (SE) reflect universalist welfare.

Figure 16. Latent Class Analysis fit indices across k

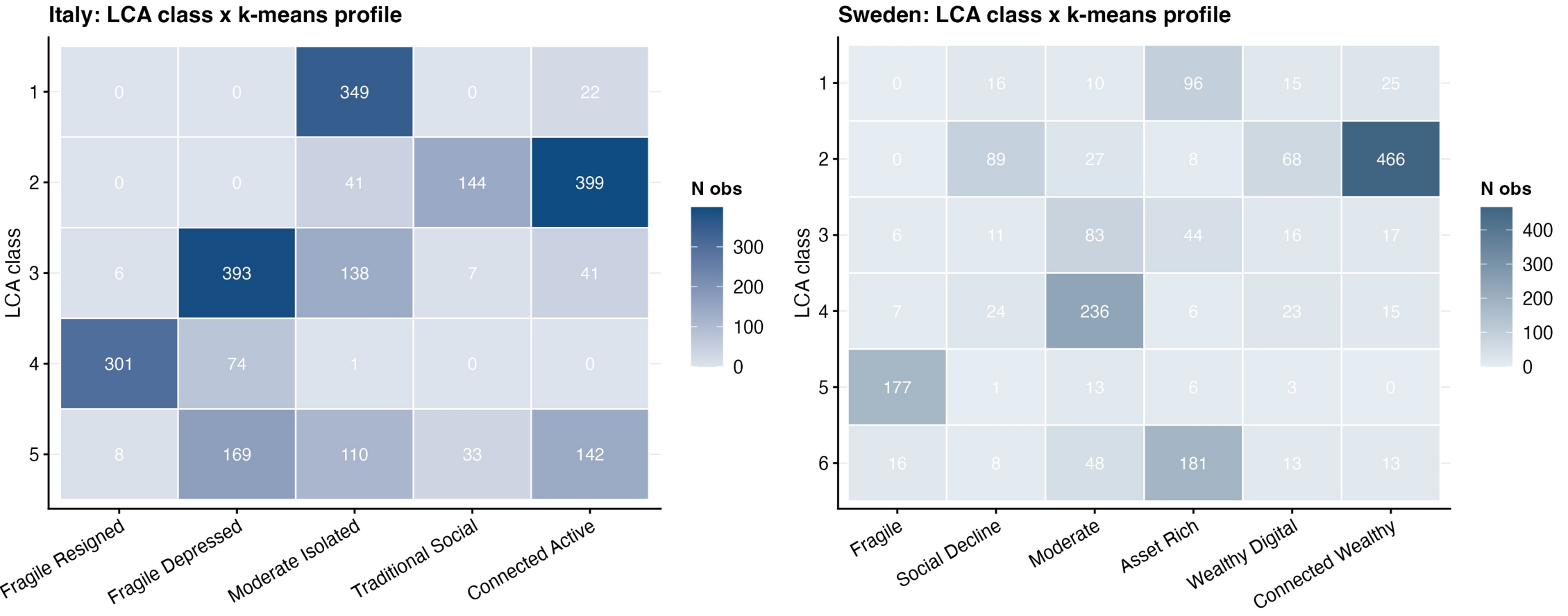
BIC, AIC, and entropy for k=2..8; k=6 chosen for cross-country comparability (dashed)



BIC-optimal k differs (IT: k=8, SE: k=4); k=6 retained on substantive/interpretability grounds.

Figure 17. Cross-method validation: LCA classes vs k-means profiles

Convergence of probabilistic (LCA) and geometric (k-means) segmentations at k=5 (IT) and k=6 (SE)

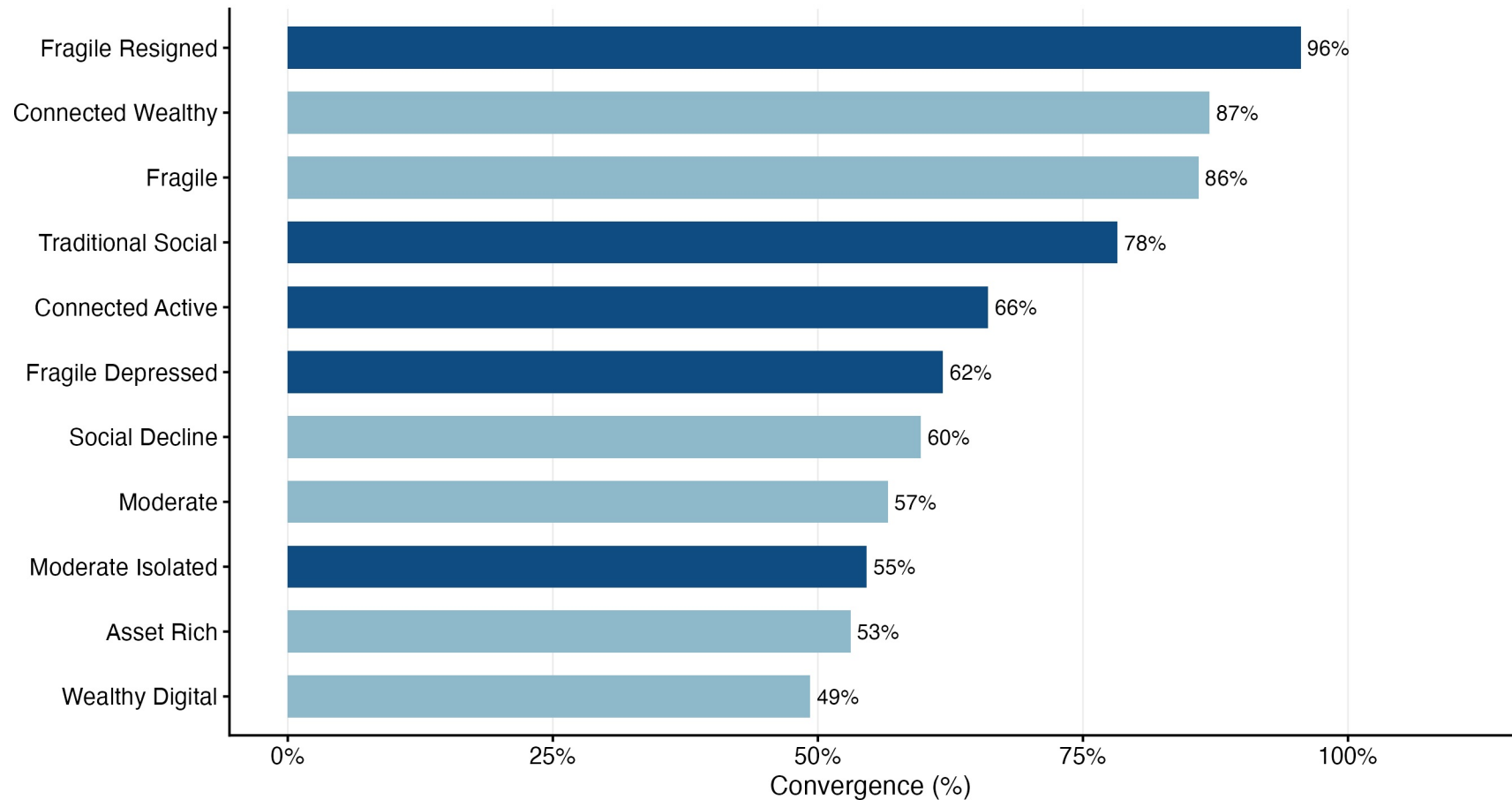


Global convergence: Italy 66.7%, Sweden 68.1%. Fragile and Connected profiles show strongest cross-method agreement.

Figure 18. Cross-method convergence by profile

Share of k-means profile members falling in the dominant LCA class

Country ■ Italia ■ Svezia

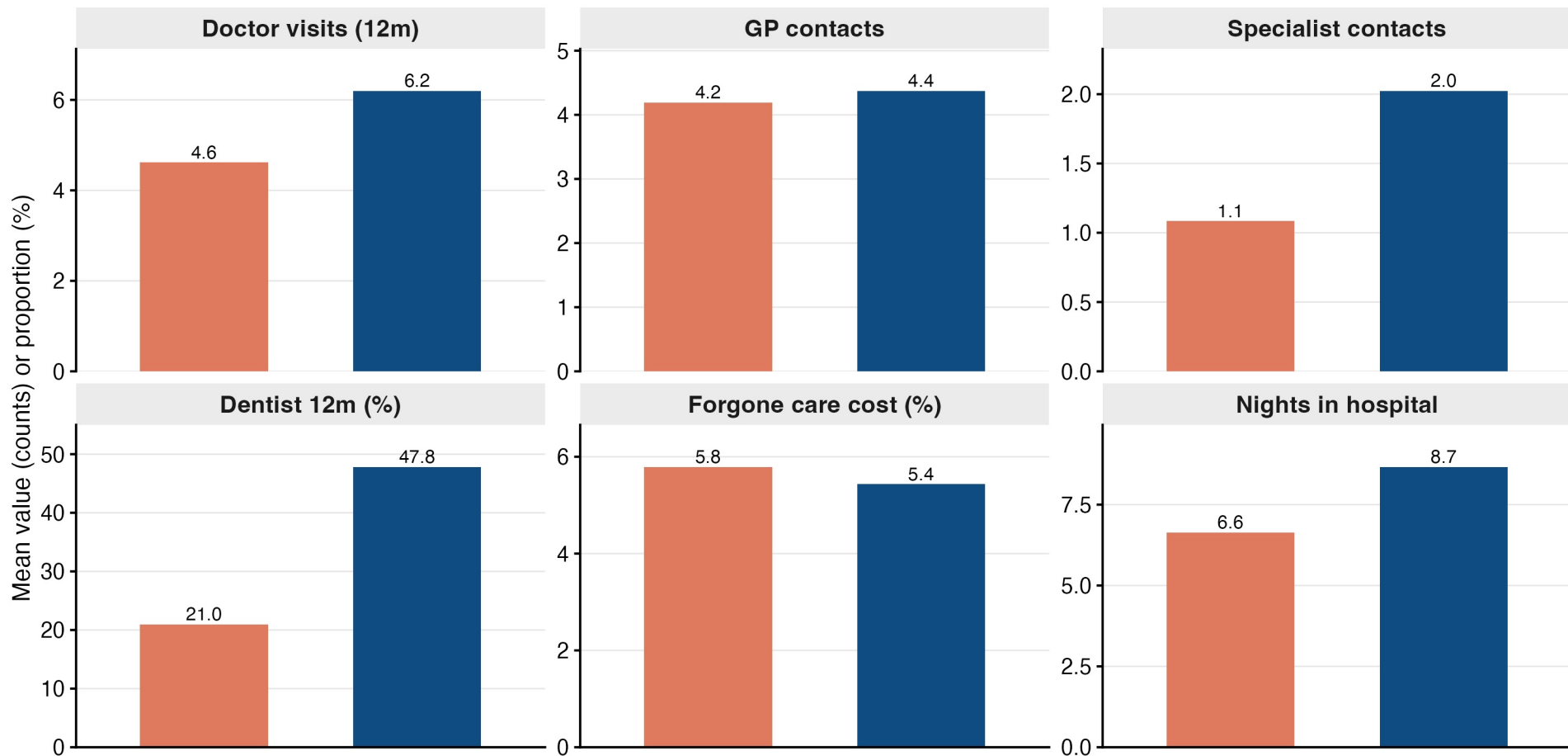


Fragile and Connected profiles show highest convergence; intermediate profiles (Wealthy Digital, Asset Rich, Moderate Isolated) are less c

Figure 19. Italian paradox: healthcare utilization, Moderate Isolated vs Traditional Social

Profiles with comparable physical conditions but different social embeddedness

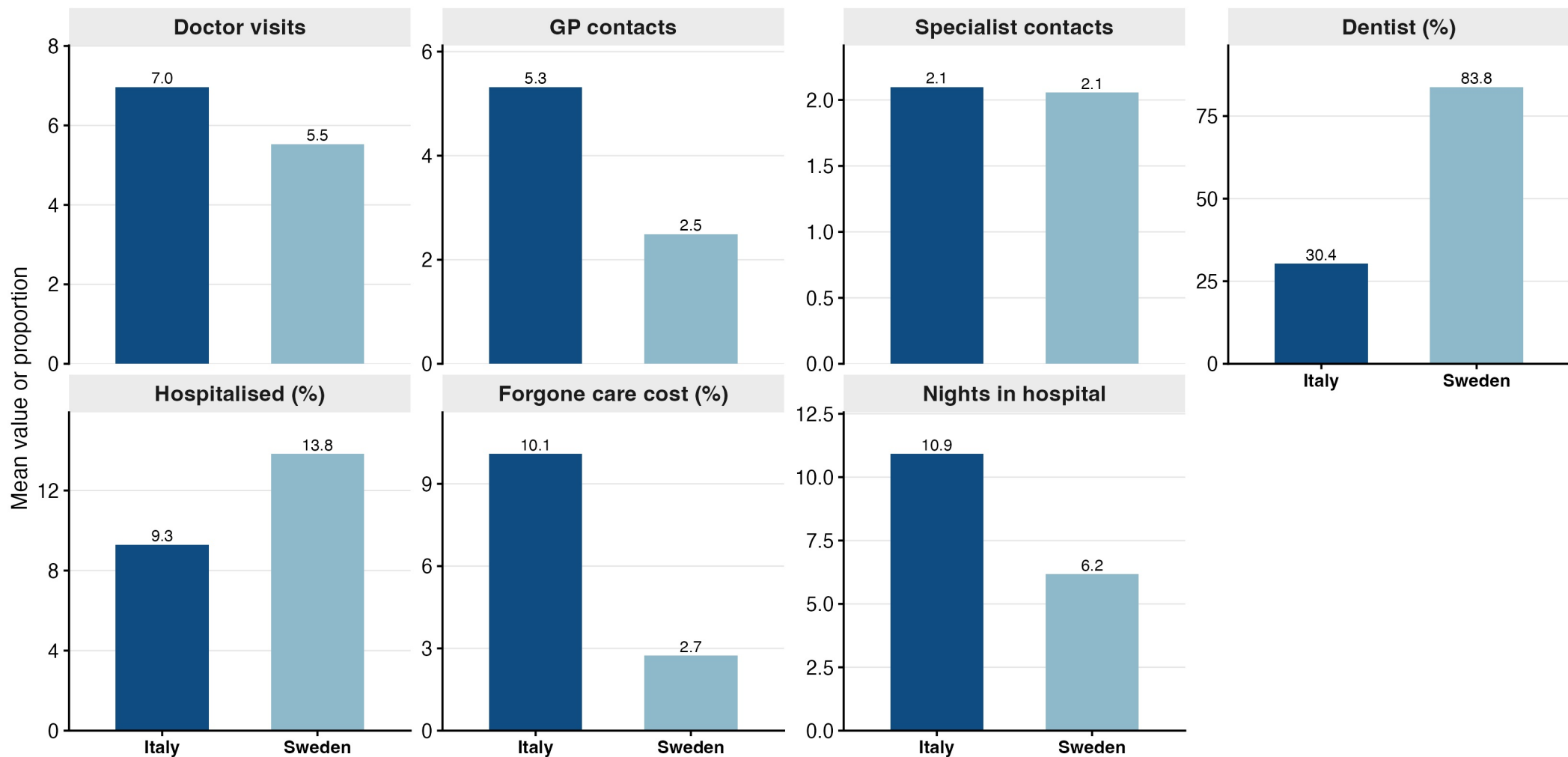
Italian profile Moderate Isolated Traditional Social



Italy, subsample with healthcare data (n=2,344). Significance: Welch t-test (continuous) or chi-squared (proportions).

Figure 20. Healthcare utilization, cross-country comparison

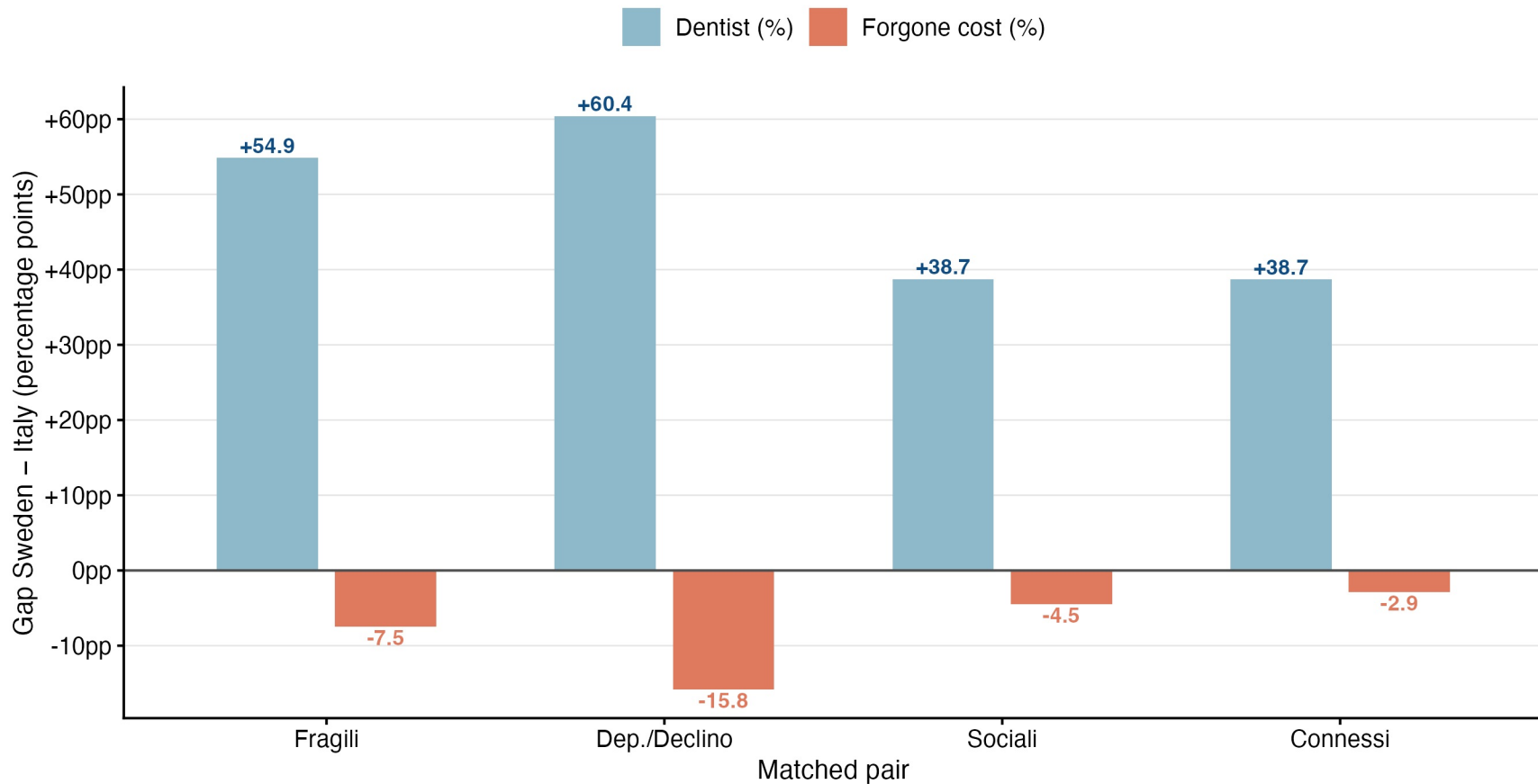
Italy and Sweden aggregate means on seven healthcare indicators



Italy n=2,344; Sweden n=1,779. Welfare regime contrast most visible on dentist access and forgone care cost.

Figure 21. Welfare regime signature: dentist access and forgone care cost

Sweden minus Italy in percentage points, across the four matched profile pairs

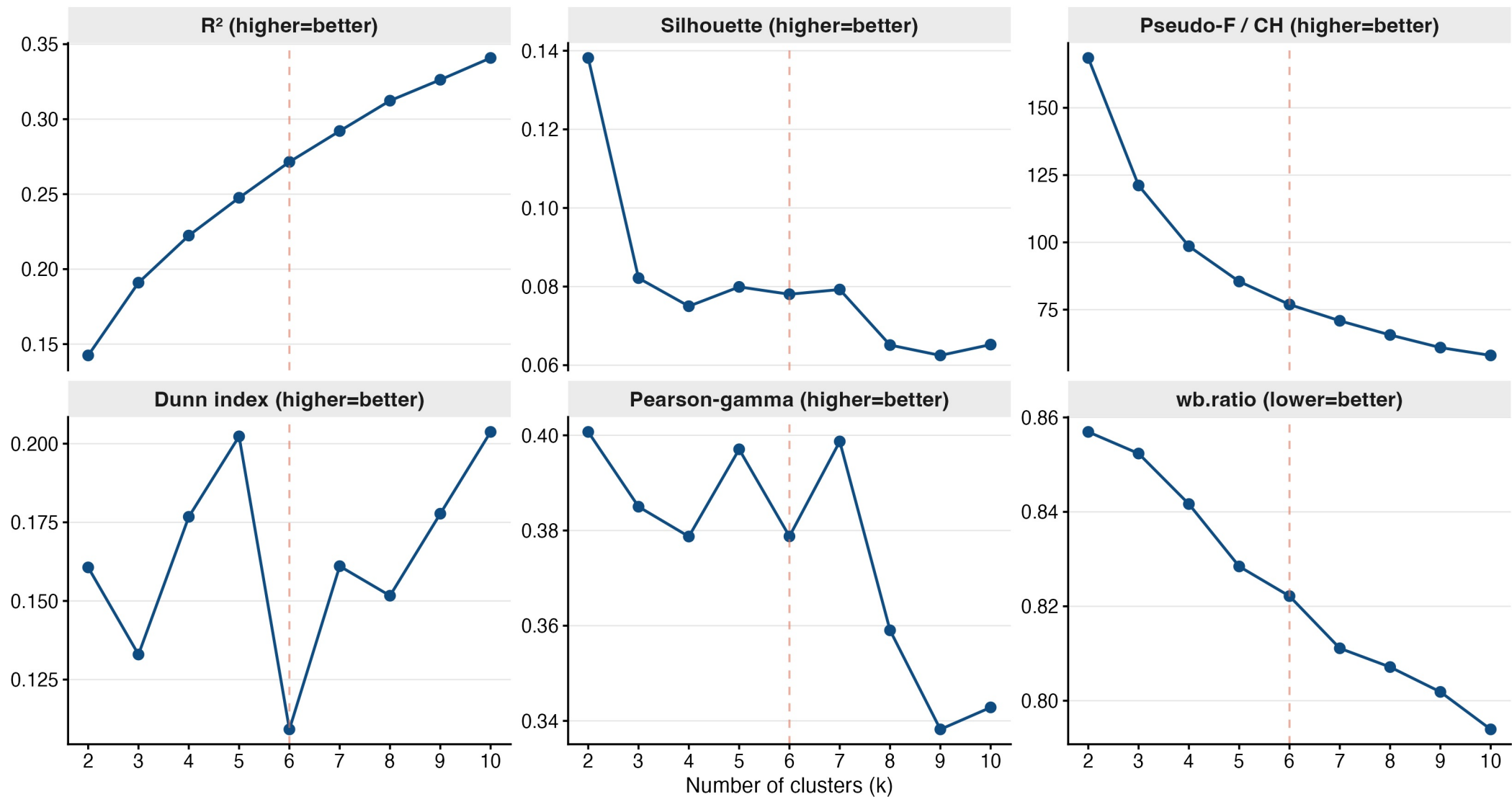


Dentist access: universal coverage gap consistently +40/+60pp in Sweden.

Forgone cost: Italian economic barrier pattern, strongest among fragile profiles.

Figure 22. Cluster quality indexes, Italy

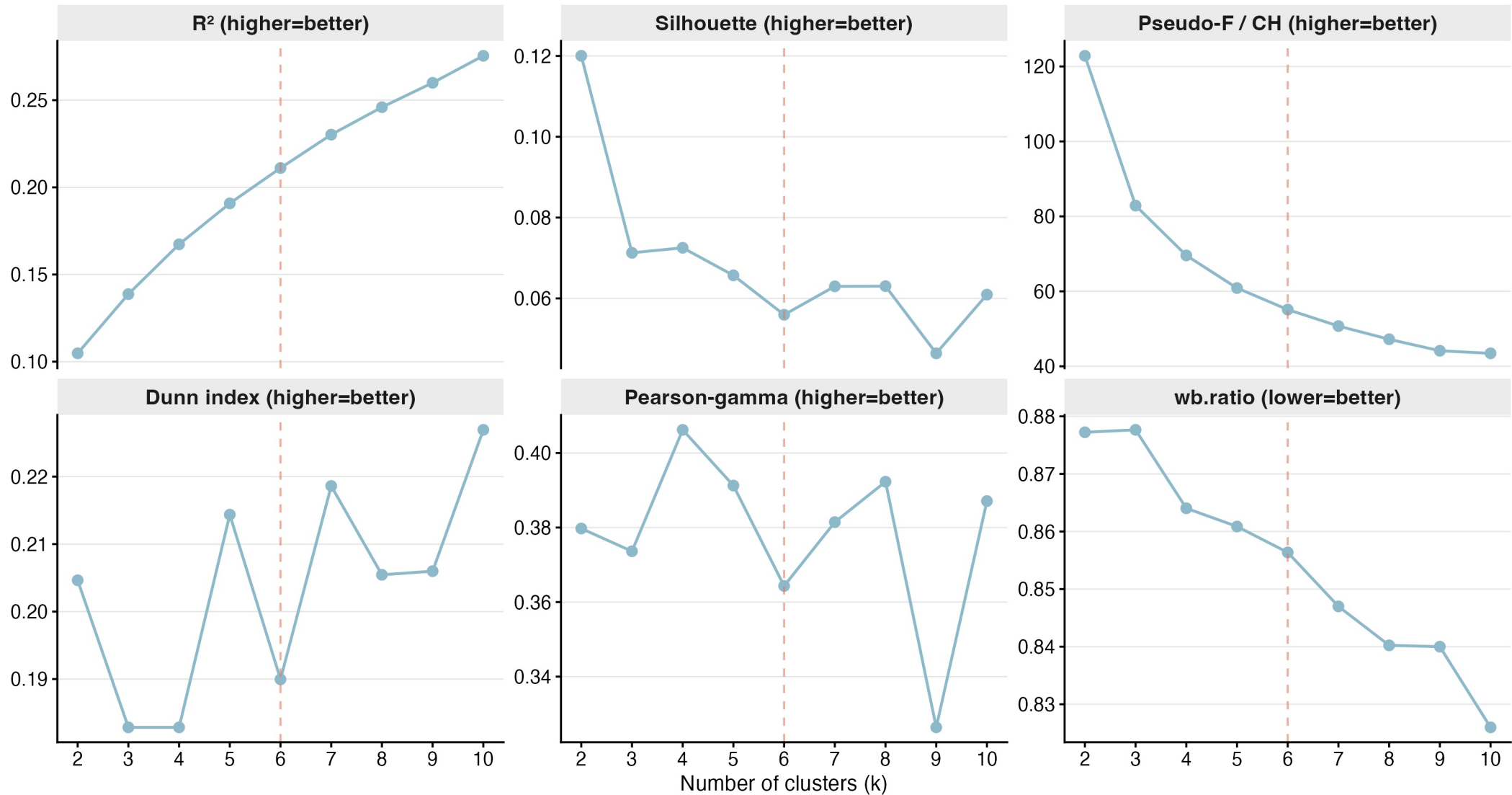
Six internal validity criteria across k ; dashed line marks $k=6$ selected



Computed on a 1,000-observation subsample for computational tractability; R^2 computed on the full sample.

Figure 23. Cluster quality indexes, Sweden

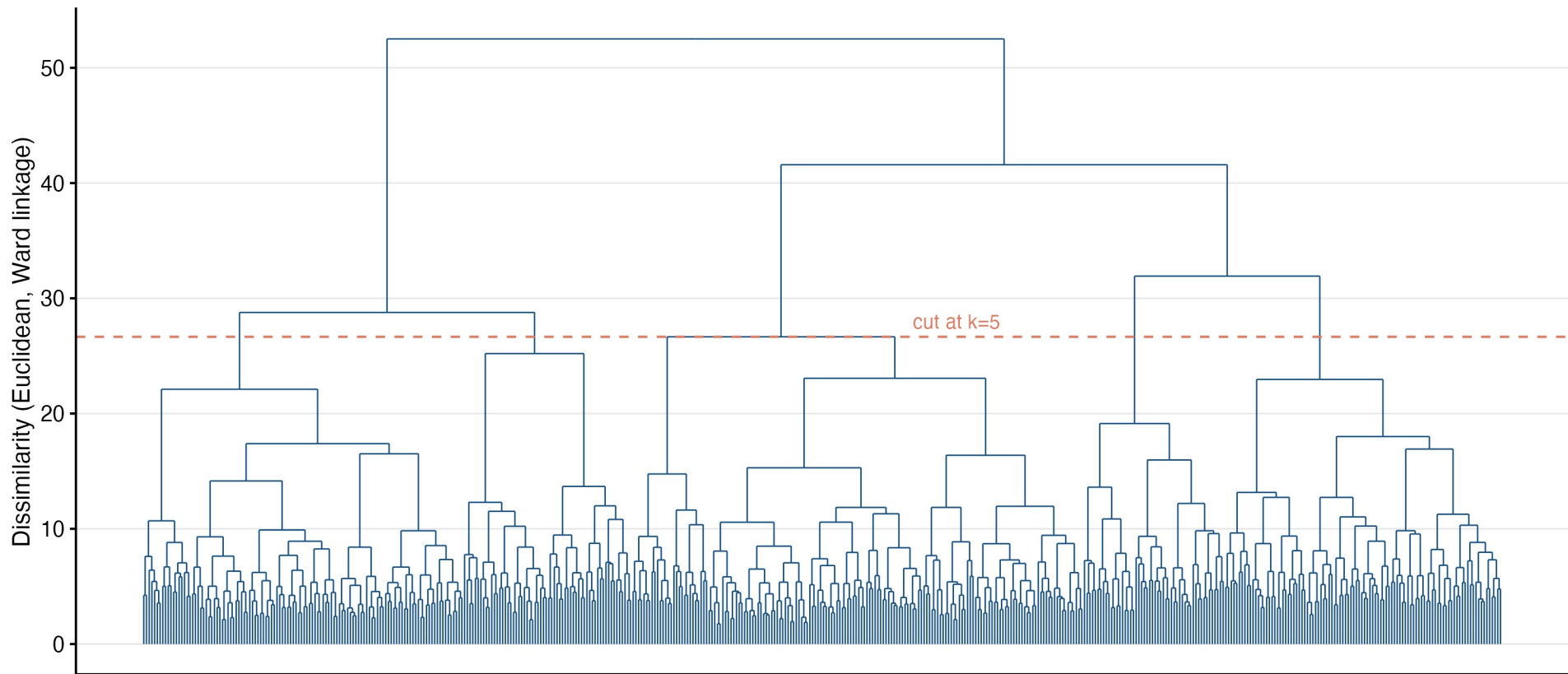
Six internal validity criteria across k ; dashed line marks $k=6$ selected



Computed on a 1,000-observation subsample for computational tractability; R^2 computed on the full sample.

Figure 24. Hierarchical dendrogram (ward.D2), Italy

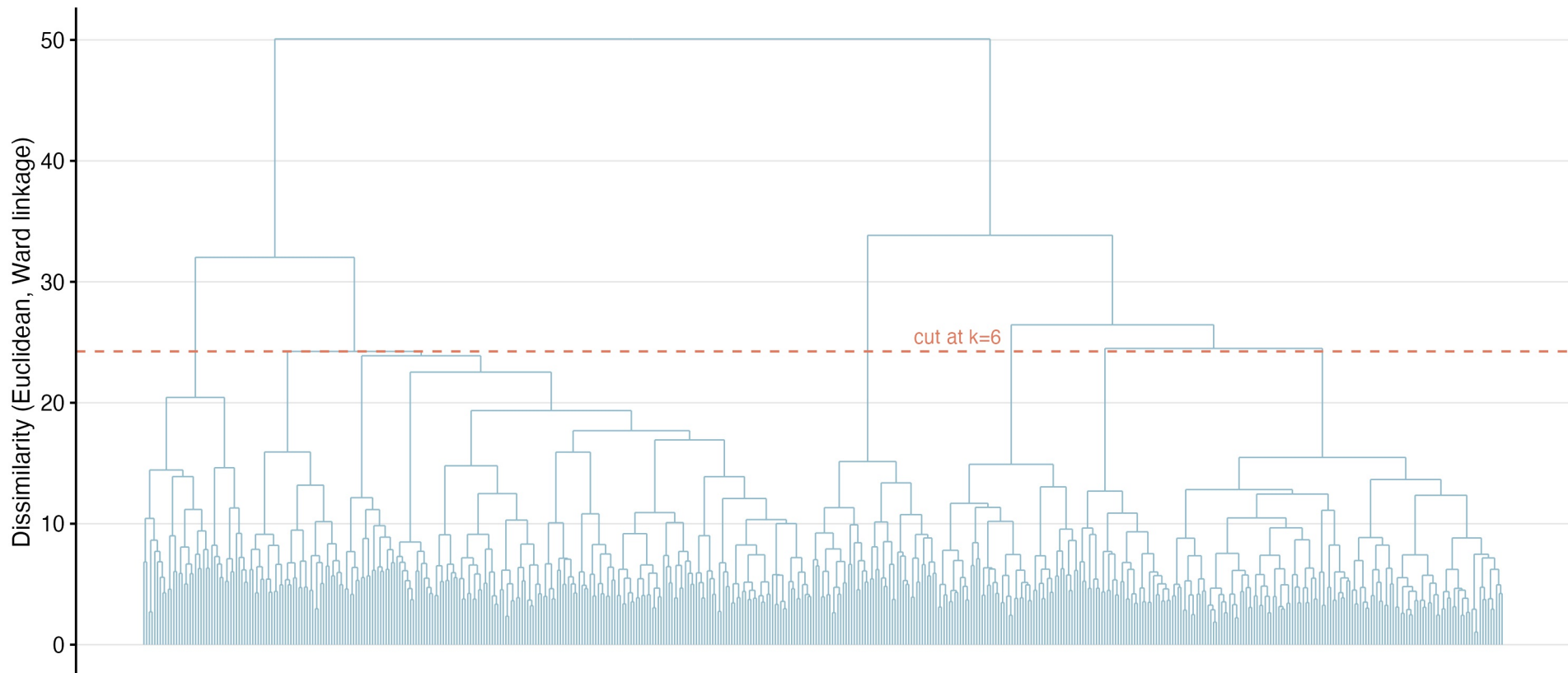
500-observation subsample; dashed line marks the k=5 cut



Hierarchical clustering reported as robustness check against k-means; see Chapter 6 for cross-method convergence.

Figure 25. Hierarchical dendrogram (ward.D2), Sweden

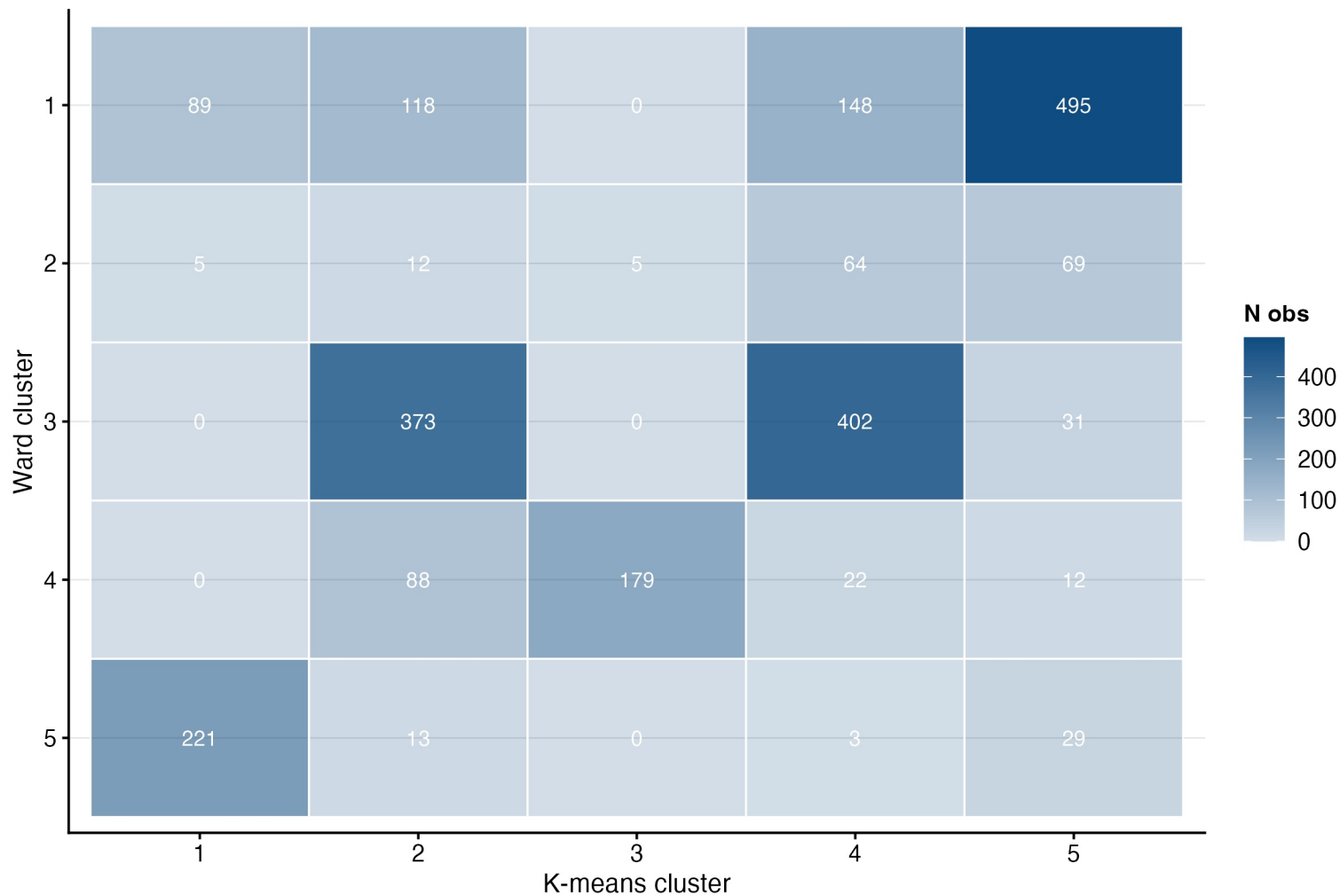
500-observation subsample; dashed line marks the k=6 cut



Hierarchical clustering reported as robustness check against k-means; see Chapter 6 for cross-method convergence.

Figure 26. K-means vs Ward hierarchical (k=5), Italy

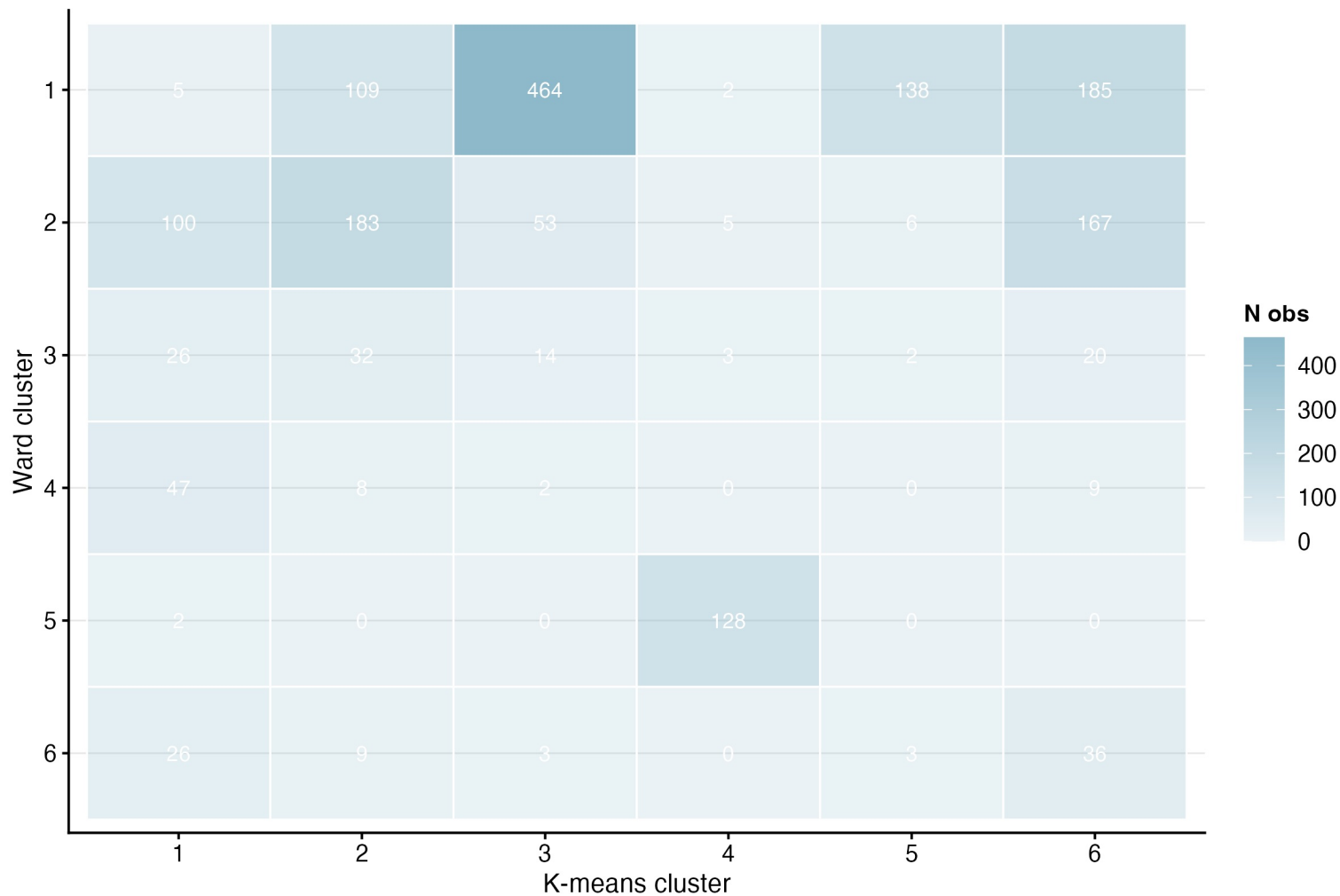
Modal-class convergence: 70.2%



Agreement measured as share of k-means members falling in the modal Ward cluster. Serves as robustness check on partition stability.

Figure 27. K-means vs Ward hierarchical (k=6), Sweden

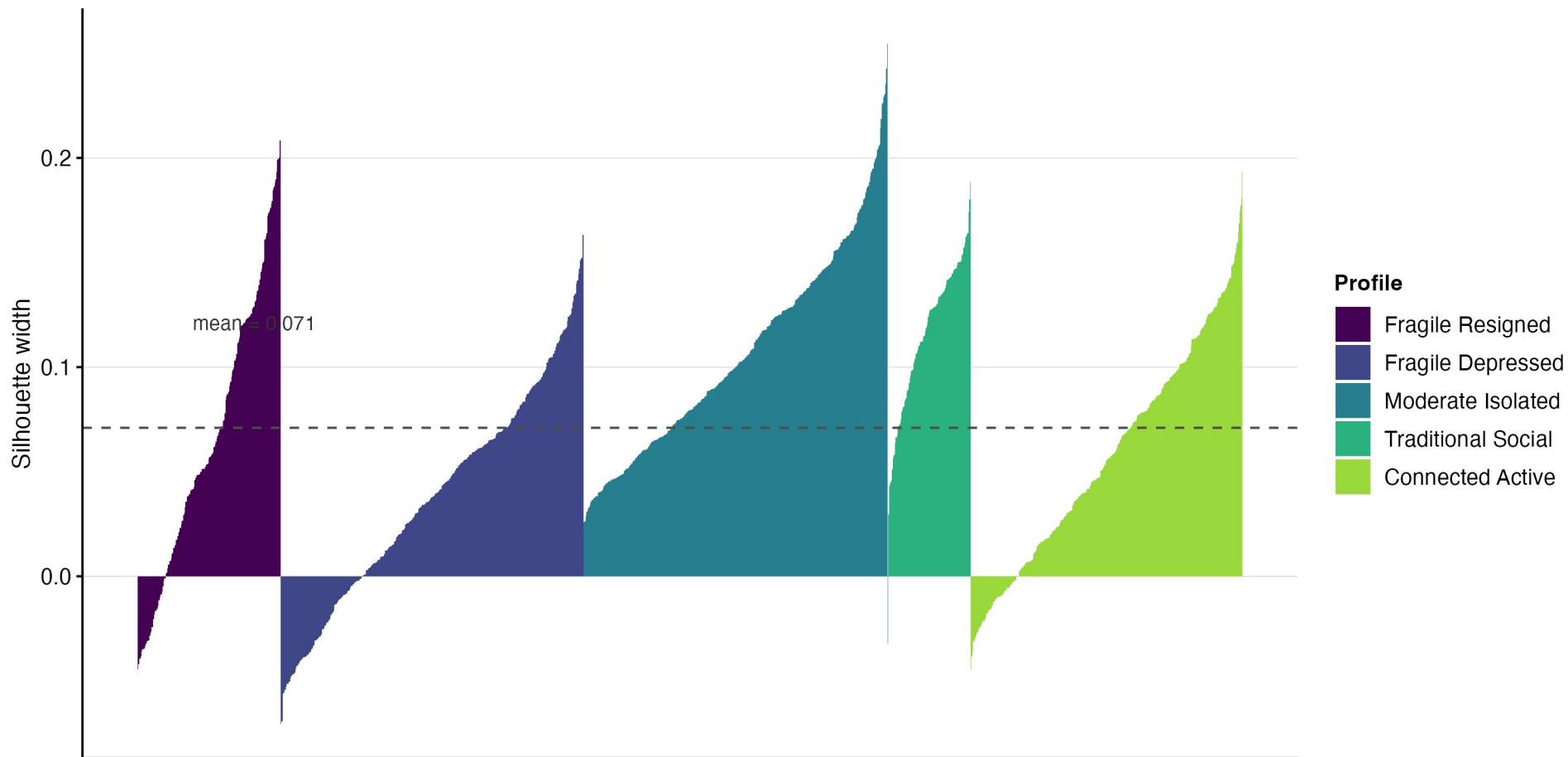
Modal-class convergence: 67.0%



Agreement measured as share of k-means members falling in the modal Ward cluster. Serves as robustness check on partition stability.

Figure 28. Silhouette width per respondent, Italy

Respondents ordered by profile and then by silhouette width



1,500-observation subsample. Positive widths indicate assignments closer to own cluster than to neighbours.

Figure 29. Silhouette width per respondent, Sweden

Respondents ordered by profile and then by silhouette width

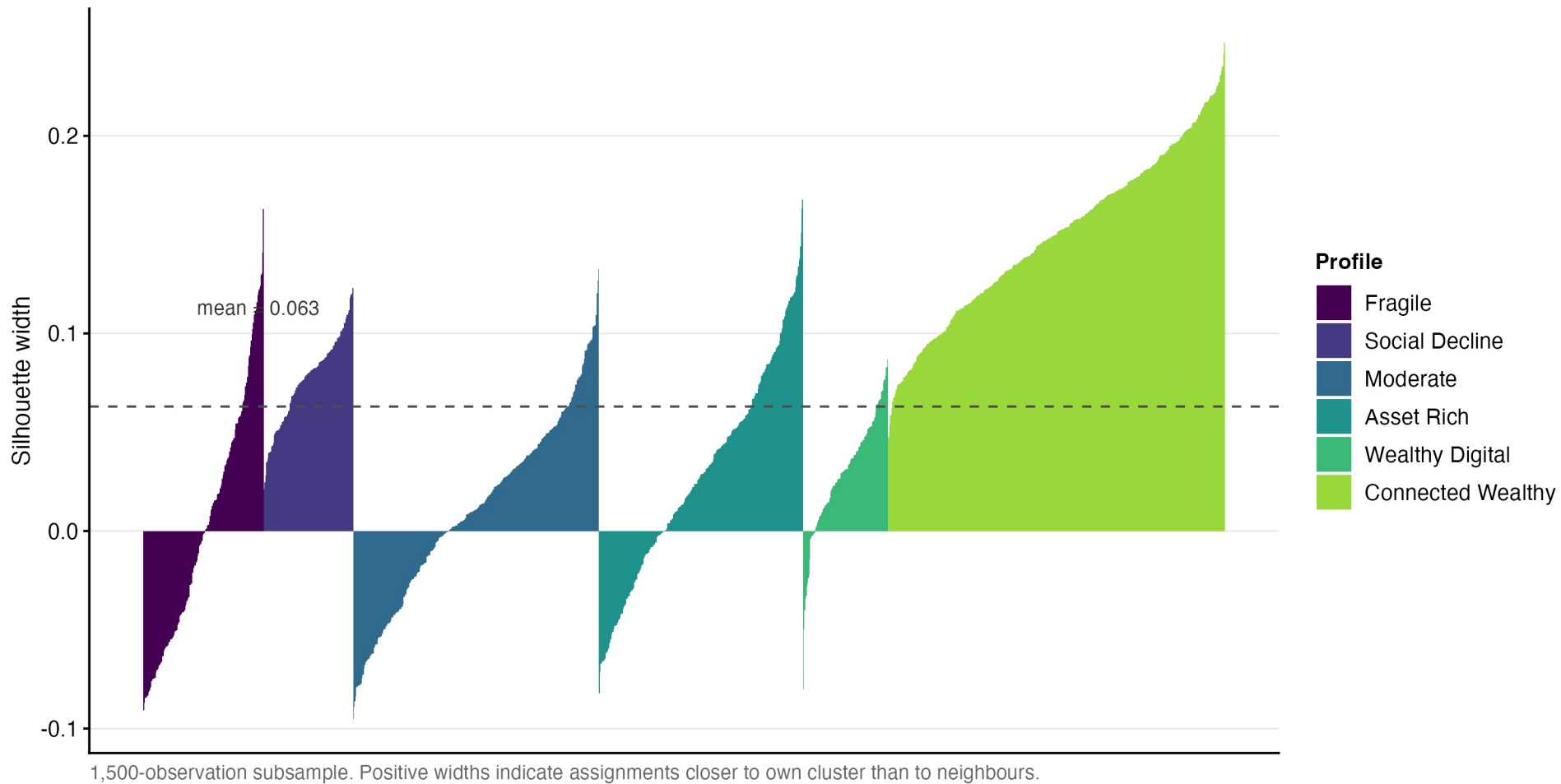
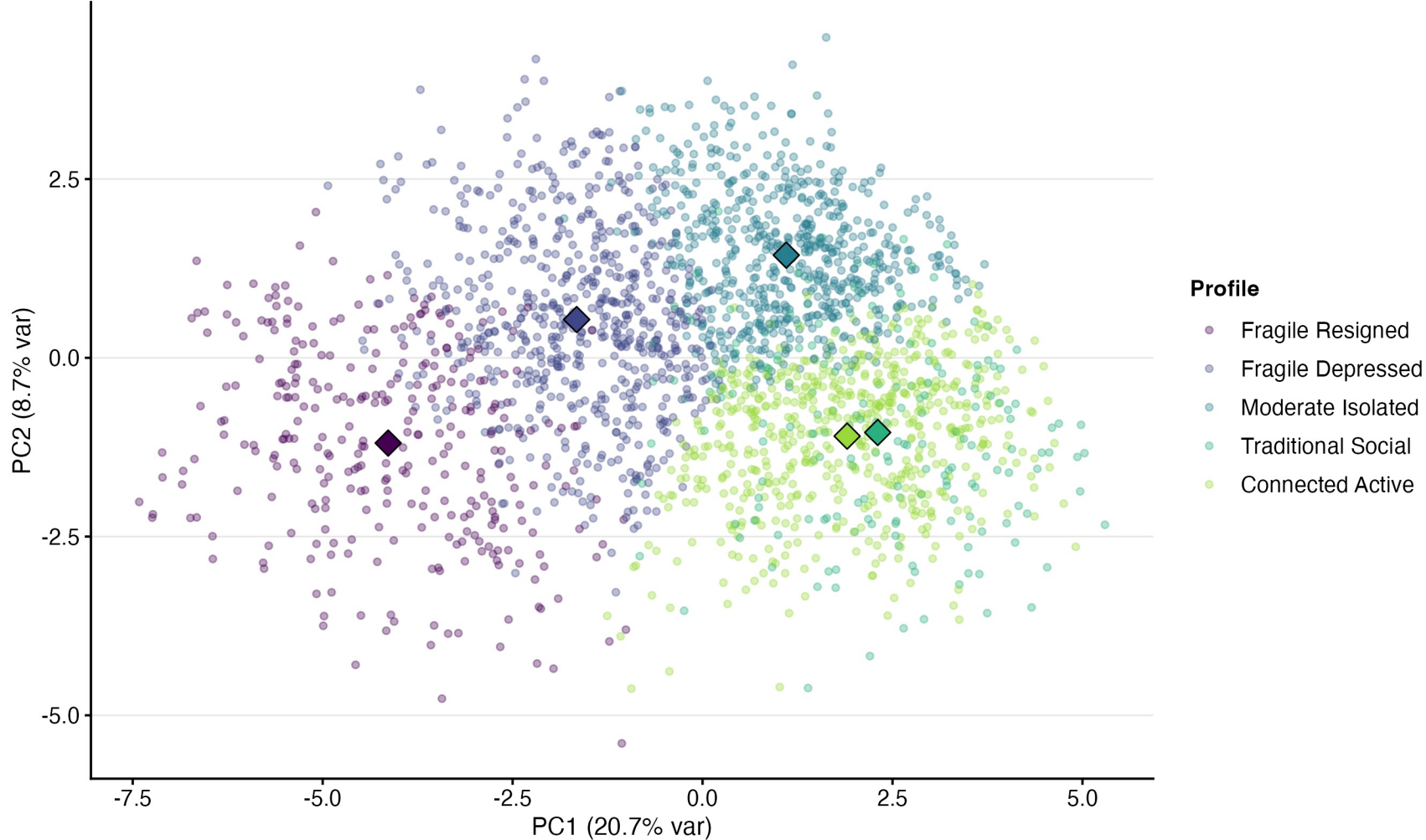


Figure 30. Respondents in first-two PC space, Italy

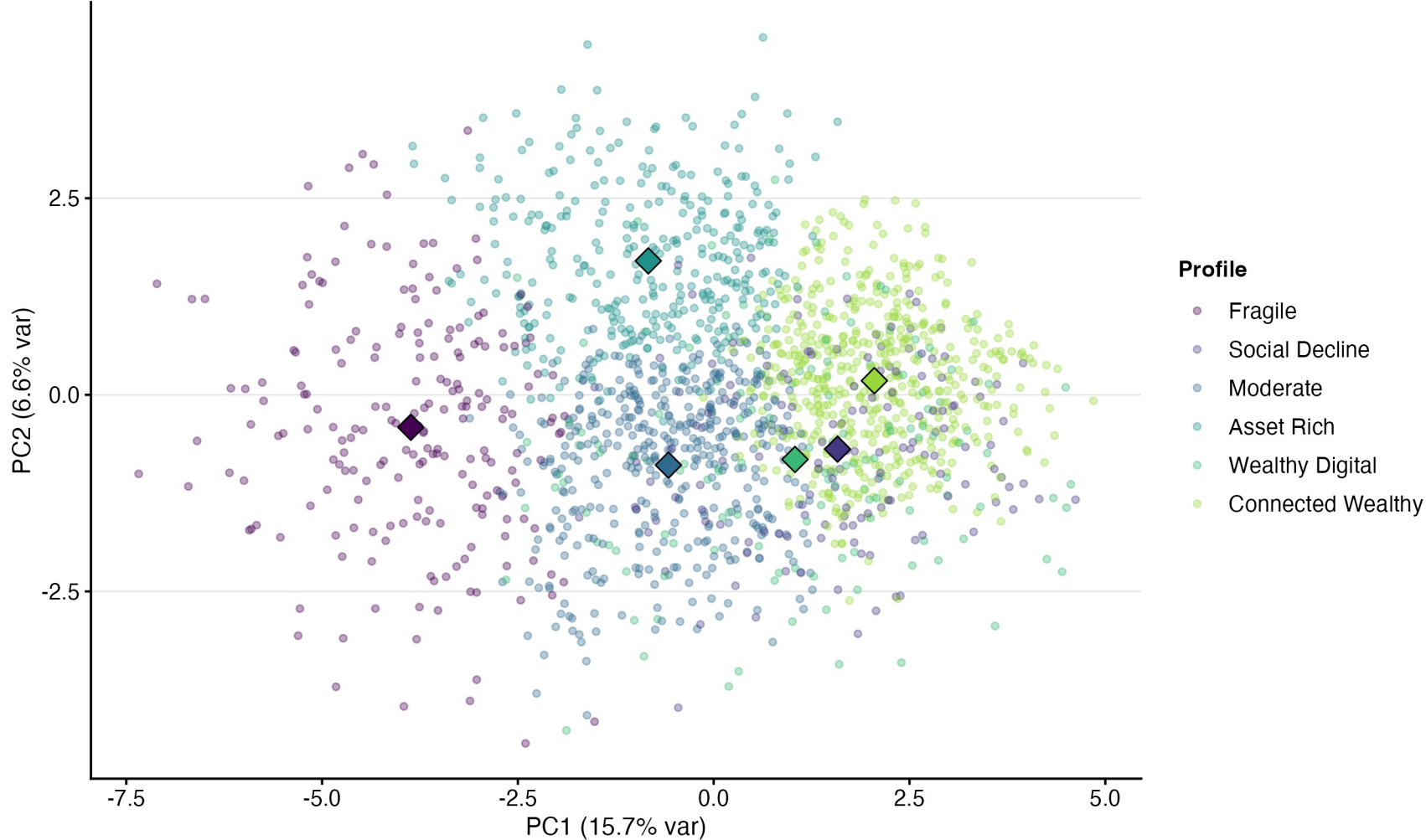
Global PCA on the analytical variable set; profiles overlaid



Diamonds = profile centroids. Separation along PC1 reflects the dominant frailty-versus-connectedness gradient.

Figure 31. Respondents in first-two PC space, Sweden

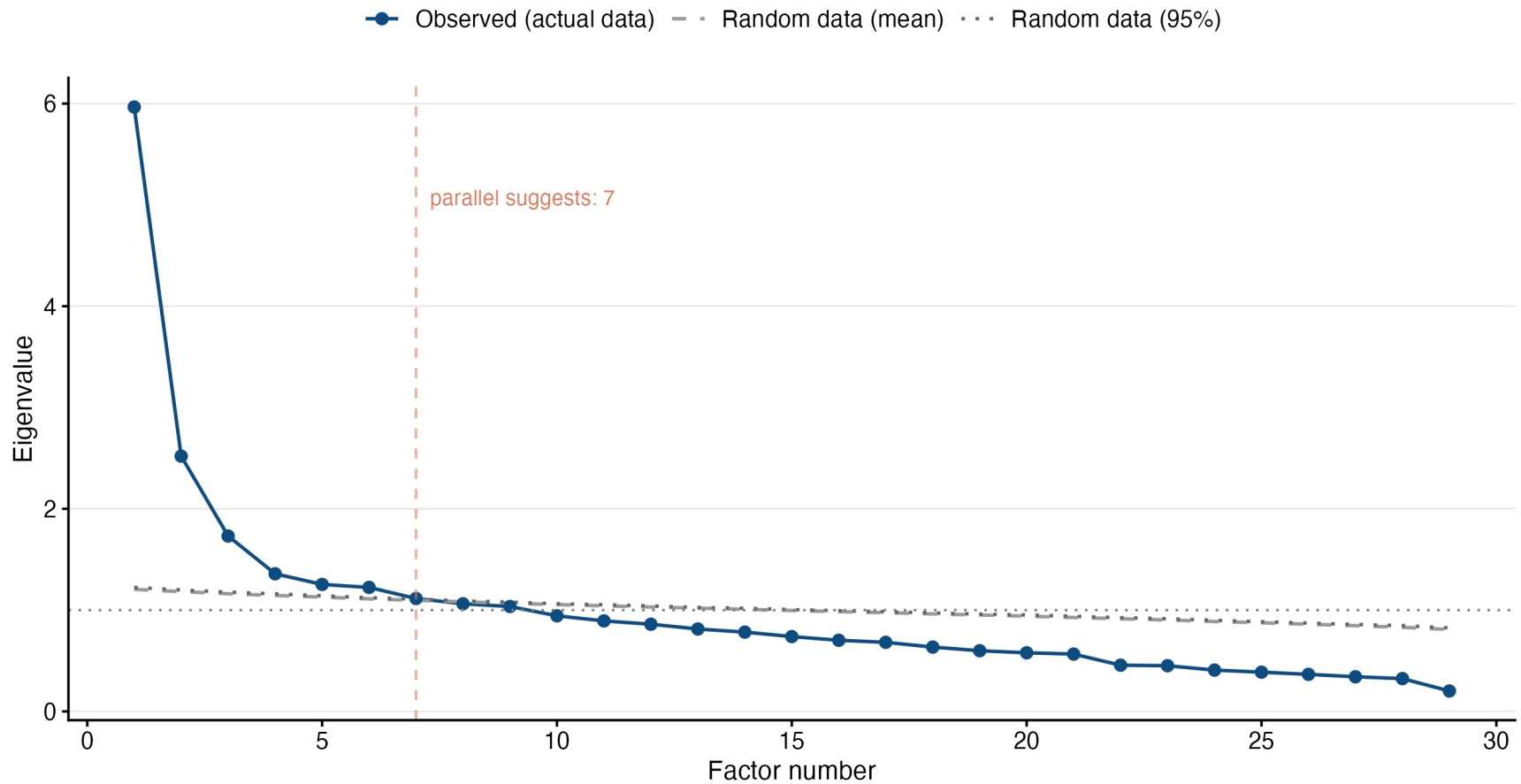
Global PCA on the analytical variable set; profiles overlaid



Diamonds = profile centroids. Separation along PC1 reflects the dominant frailty-versus-connectedness gradient.

Figure 32. Parallel analysis, Italy

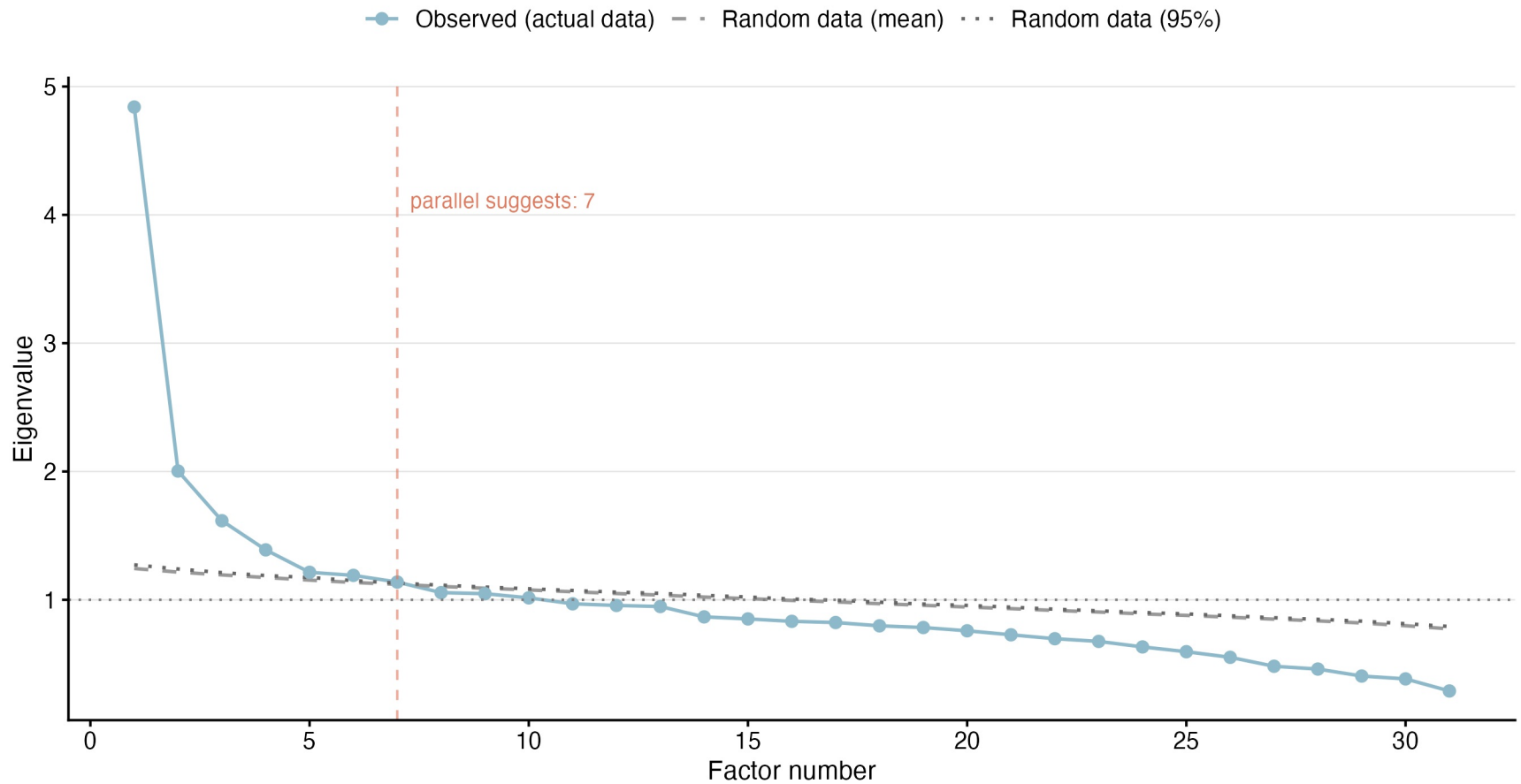
Observed eigenvalues against mean and 95% quantile of eigenvalues from random matrices



Parallel analysis with 50 random datasets. Suggested factors: 7 (crosses the 95% random threshold).

Figure 33. Parallel analysis, Sweden

Observed eigenvalues against mean and 95% quantile of eigenvalues from random matrices



Parallel analysis with 50 random datasets. Suggested factors: 7 (crosses the 95% random threshold).

Figure 34. Communalities per variable, Italy and Sweden

Extracted from the 6-factor principal-axis solution with varimax rotation

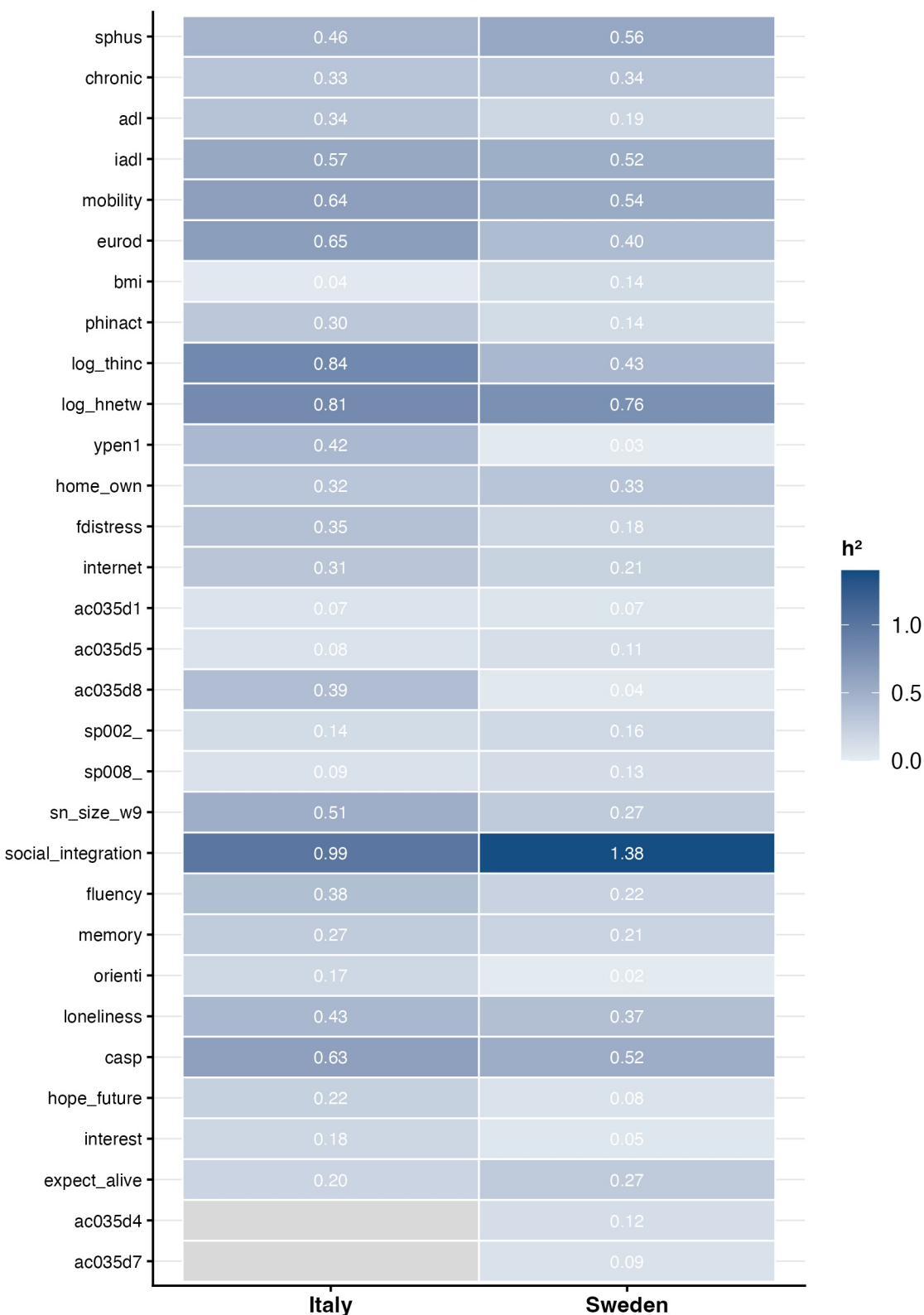
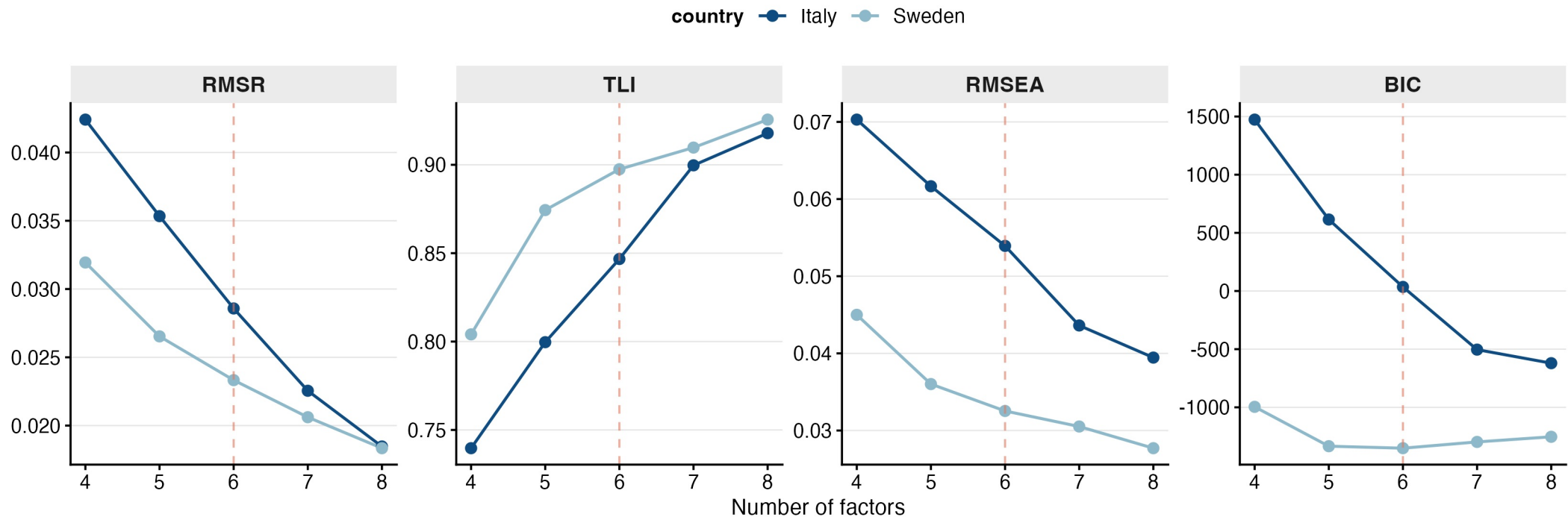
Communalities (h^2) summarize shared variance captured by the factor solution. Values > 1

Figure 35. Factor-analysis fit indices across number of factors

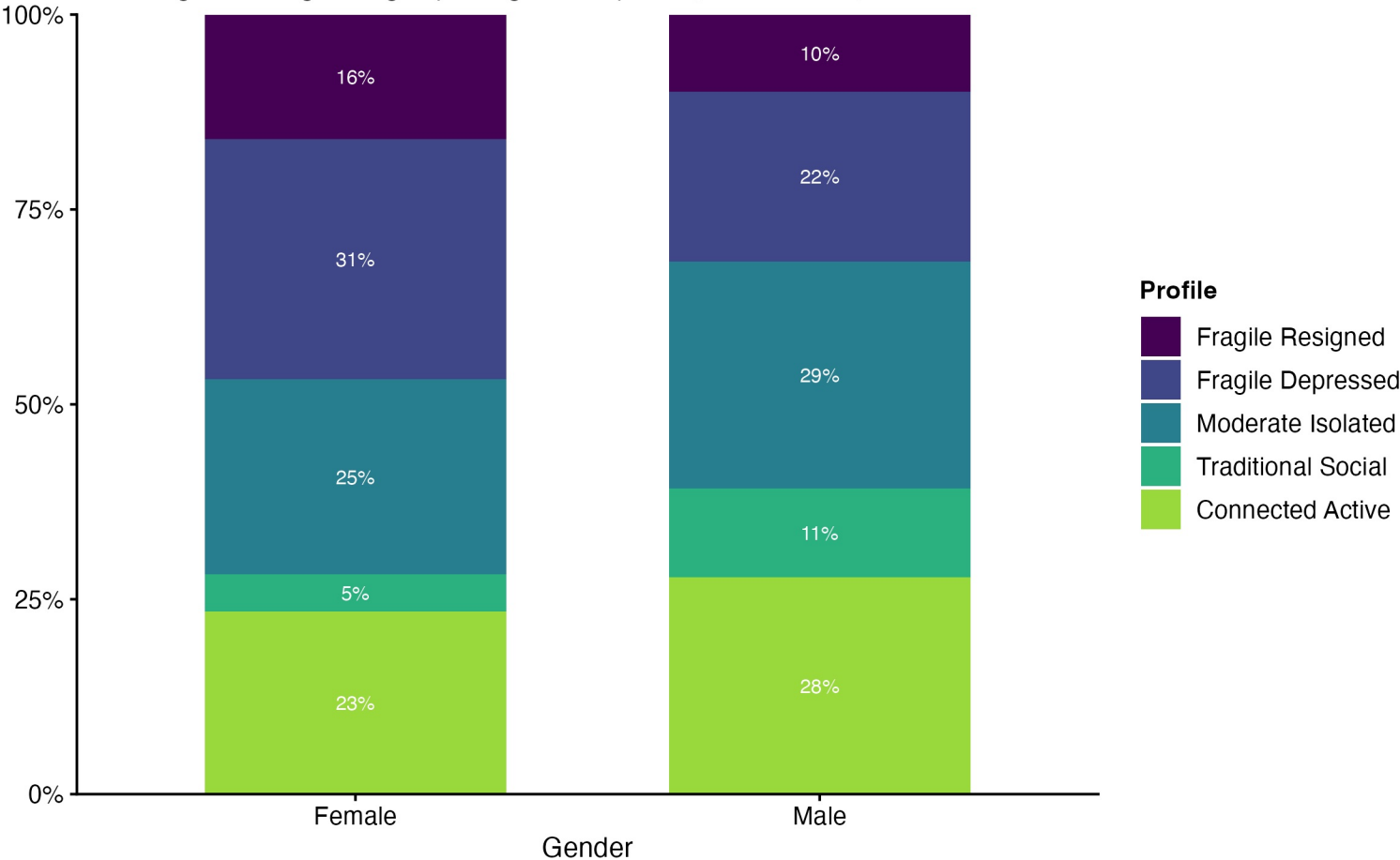
Four fit indices for $k = 4$ to 8 factors; dashed line marks the $k=6$ design choice



The six-factor solution is conservative relative to BIC optima ($k=7$ IT, $k=4$ SE) but is retained for cross-country comparability.

Figure 36. Profile composition by gender, Italy

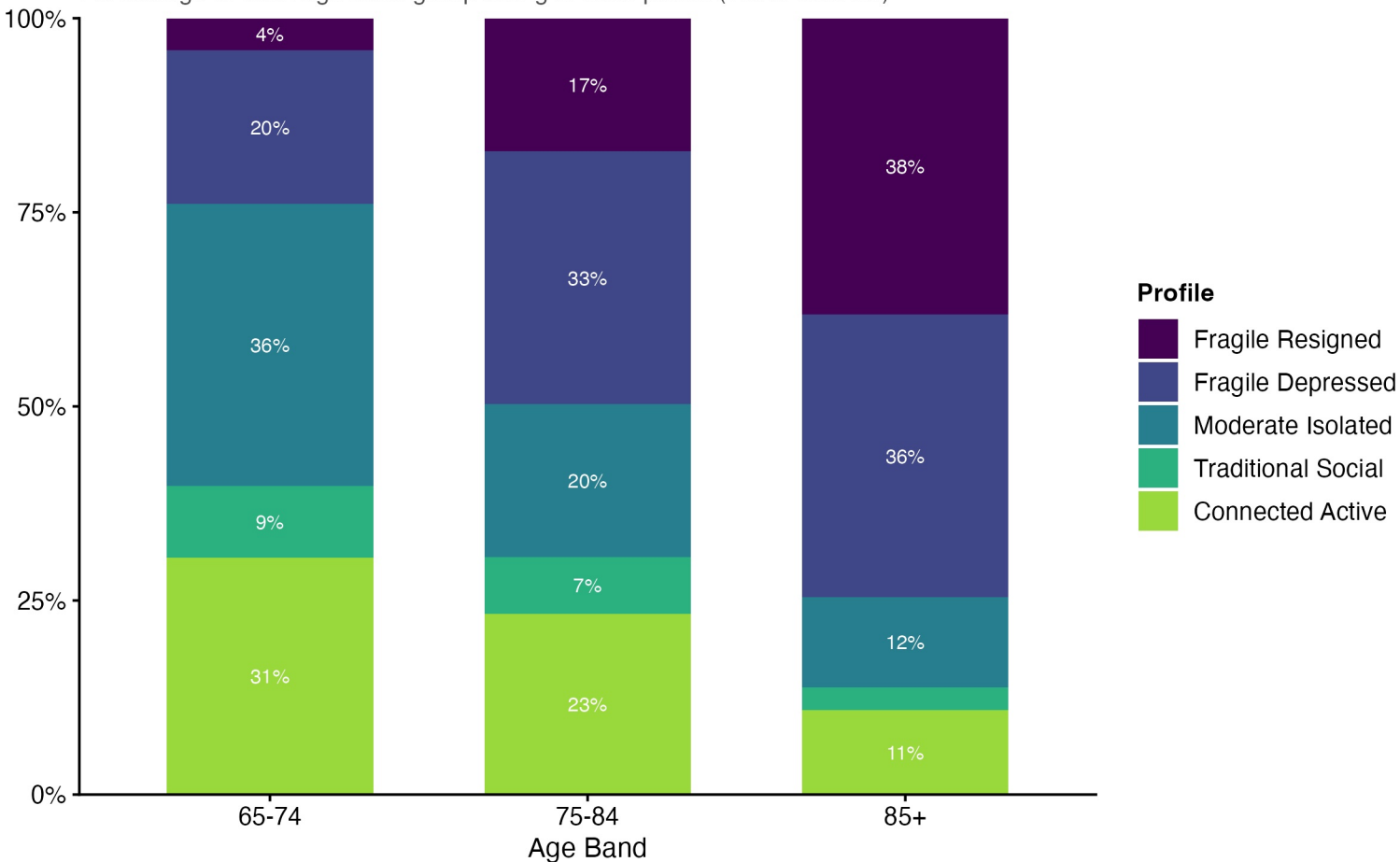
Percentage of each gender group falling in each profile (100% stacked)



Percentages within each column sum to 100%.

Figure 37. Profile composition by age band, Italy

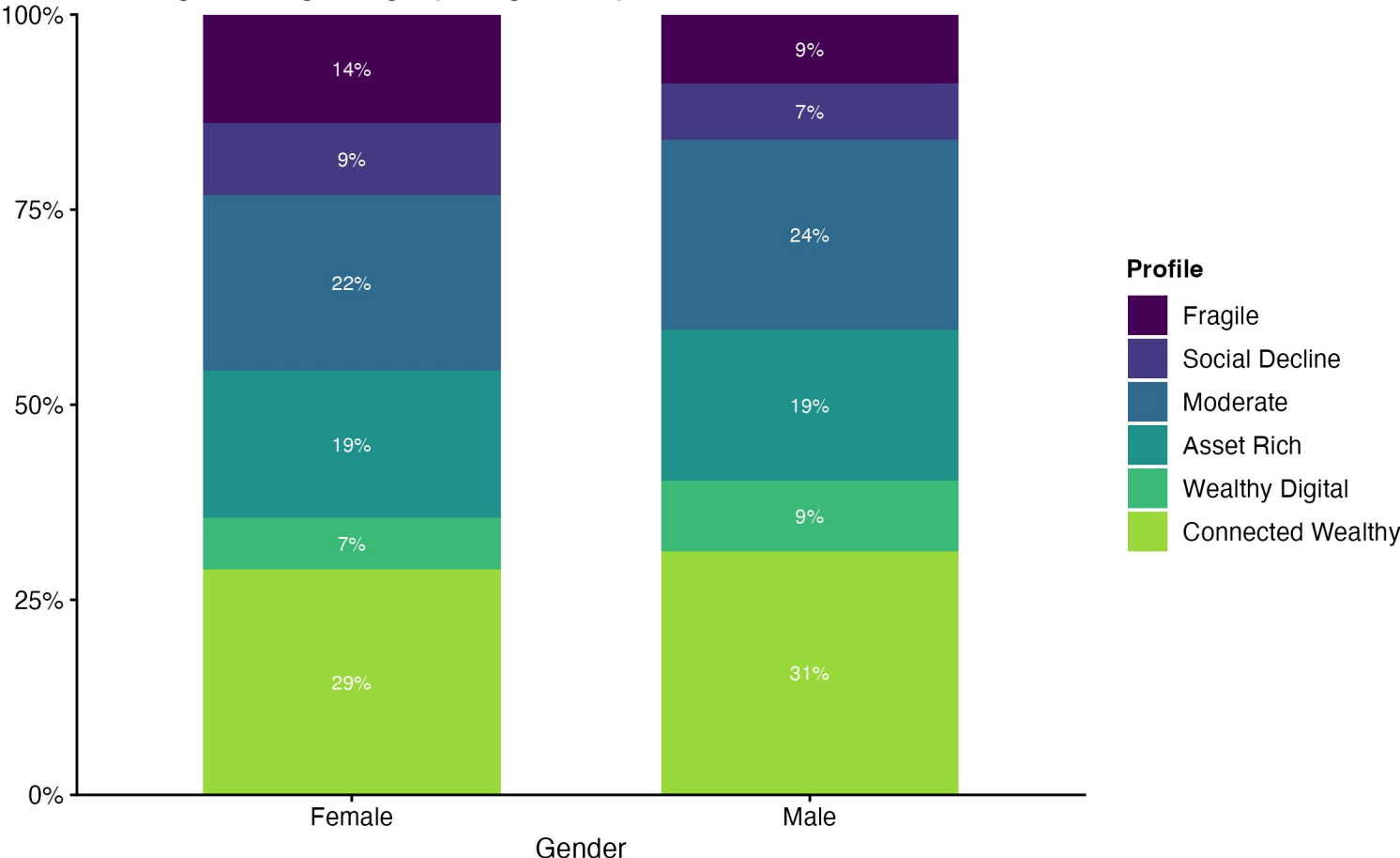
Percentage of each age band group falling in each profile (100% stacked)



Percentages within each column sum to 100%.

Figure 38. Profile composition by gender, Sweden

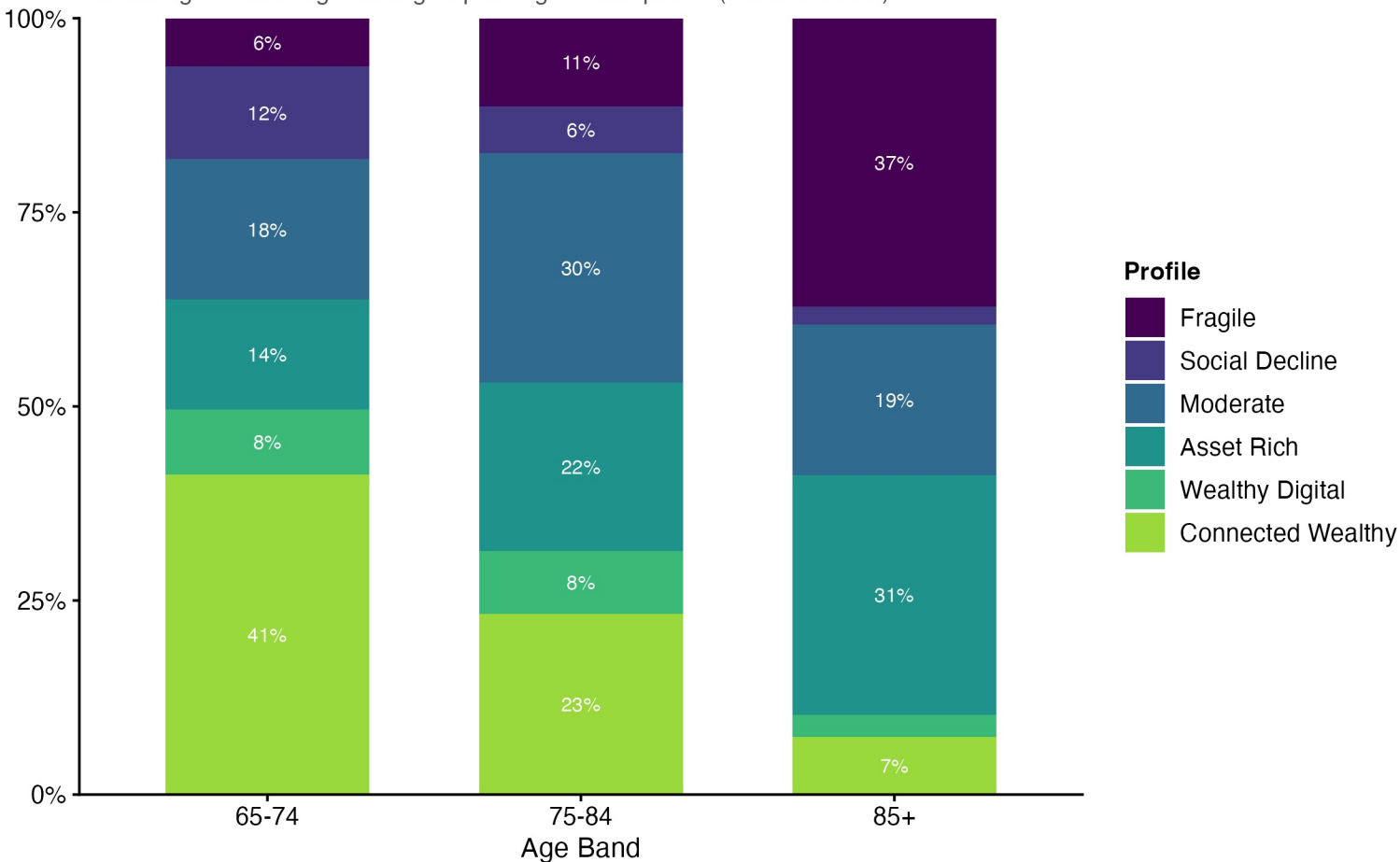
Percentage of each gender group falling in each profile (100% stacked)



Percentages within each column sum to 100%.

Figure 39. Profile composition by age band, Sweden

Percentage of each age band group falling in each profile (100% stacked)



Percentages within each column sum to 100%.